

CLIMATE CATALYSTS

## CATALYST OF CHANGE

Navigating Climate Change Together

"Catalysts of Change: Navigating Climate Challenges Together" is a comprehensive initiative aimed at charting a course towards a diversified and sustainable future for our planet. Rooted in the imperative of preserving the health and resilience of our ecosystem, this program is designed to secure benefits not only for the present generation but also for those to come. By leveraging a wealth of best practices, innovative approaches, climate action projects, and scientific research, the initiative offers a multifaceted strategy to combat climate change from various perspectives. It brings together diverse stakeholders to collaborate on solutions that address the complex challenges posed by climate change.

One of the key features of "Catalysts of Change" is its emphasis on the involvement of young people from around the world. Recognizing the importance of fresh perspectives and innovative ideas, the initiative has actively engaged youth from countries such as Indonesia, Nigeria, Bangladesh, Iran, the United States, and the United Kingdom. These young voices bring a wealth of diversity and creativity to the table, enriching the discourse on climate resilience and sustainability. By empowering youth to take an active role in shaping the future, the initiative aligns with the objectives outlined in the Agenda 2030.

Overall, "Catalysts of Change" presents a unique opportunity to empower tomorrow's advocates and work towards achieving the UN Sustainable Development Goals at all levels, from grassroots to global. By fostering collaboration, innovation, and action, this initiative enables individuals and communities to play an active role in addressing climate change and promoting sustainable development. Whether it is through local grassroots initiatives or larger-scale global efforts, this book empowers people to make a positive impact and contribute to a more sustainable future for our planet. Together, we can work towards achieving the Sustainable Development Goals and creating a world where everyone can thrive within planetary boundaries.



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The climate is changing, but why aren't we?



# **CATALYST OF CHANGE**

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Navigating Climate Change Together

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# CATALYST OF CHANGE

Navigating Climate Change Together

## Climate Catalyst

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## Foreword 1: Rizky Utami, S.E., M.Acc.Fin

In a world where the urgency of climate change demands collective action, "Catalyst of Change: Navigating Climate Change Together" stands as a beacon of hope and guidance. As a supervisor, it's my privilege to introduce this insightful book, which not only highlights the pressing issues we face but also offers practical solutions and a roadmap for a sustainable future.

Through its pages, readers will discover the power of collaboration, innovation, and determination in tackling the greatest challenge of our time. Every contributor has woven together a tapestry of insights, strategies, and narratives that inspire hope and action. From the halls of academia to the frontlines of environmental activism, this book represents a convergence of diverse perspectives and expertise, united by a common goal: to catalyze meaningful change.

But perhaps most importantly, "Catalyst of Change" underscores the power of collaboration. It reminds us that no single individual, organization, or nation can address climate change alone. Instead, it is through partnership and solidarity that we can forge a path forward.

As you embark on this journey through "Catalyst of Change," I encourage you to approach its contents with an open mind and a determination to be part of the solution. For it is only by working together, across borders and boundaries, that we can navigate the challenges of climate change and build a more sustainable and equitable world for future generations.

## Foreword 2: Samintang



In today's world, young people find themselves at the forefront of a crisis they did not create: climate change. Despite bearing the least responsibility for the environmental degradation that plagues our planet, they are among the most severely impacted by its consequences. It is this profound injustice that serves as the impetus behind the creation of Catalyst of Change: Navigating Climate Challenges Together.

Within the pages of this book, we confront the stark reality of our changing climate with a sense of urgency and determination. We acknowledge the pressing need for collaborative efforts and innovative solutions to combat the multifaceted challenges posed by climate change. Through the diverse perspectives of young activists, scientists, and visionaries, Catalyst of Change offers a comprehensive exploration of the complex issues at hand, coupled with practical strategies for meaningful action.

As we embark on this collective journey, let us recognize the power that resides within each of us to effect change. Let us draw inspiration from the resilience and passion of young leaders worldwide, and let us commit ourselves to becoming catalysts for positive transformation. Together, we can forge a path towards a more sustainable, just, and equitable future for all.



### Foreword 3: Kwan Wirawan Kwandou



In recent times, the issue of climate change has become a significant global concern, impacting various aspects of human life beyond just the environment. The eastern part of Indonesia, also known as Kawasan Timur Indonesia, is abundant in natural resources. However, this region is also extremely vulnerable to the adverse effects of climate change due to recent business activities that have severely impacted and injured our environment.

Furthermore, the involvement and awareness of young people regarding climate change and environmental damage are still very minimal. Climate Catalyst strives to fill this gap by presenting various efforts to increase active participation and awareness among young people in Makassar through both formal and non-formal education.

This book represents our collective effort, featuring a diverse range of ideas and insights from youths in South Sulawesi, as well as from individuals worldwide, including students, lecturers, and researchers from different countries and universities. We hope that readers will find this book both informative and inspiring and that it encourages them to join us in addressing climate change and promoting a more sustainable future.

## Foreword 4: Nurul Habaib Al Mukarammah



The climate has been observed as a topic only able to be influenced through high-political discussions and negotiations which roadblocks various groups to meaningfully participate, particularly the youth. Often, climate policies and laws disregards the reality, the voices in the grassroots, which lead to such a policy failing to solve the root cause and unable to be translated in local context. Therefore, Climate Catalysts always put a high ambition to break those elitist boundaries by proliferating a space for youth to learn, grow and connect with fellow climate activists to advocate for and influence climate policy. Those ambitions include the initiation of this book, where youth collectively contemplate and seek solutions to navigate climate challenges.

From global north to south, this book serves as a landmark for youth around the globe to voice out their critical concerns and innovative solutions for climate. It aims to enlighten readers by providing a portrayal of real life environmental catastrophe and climate change impacts on the ground, along with plausible solutions. At the end, we aspire for this book to become an avenue of youth to influence climate actions and policies, putting the voice of youth as a primary matter.

One idea may bring impact, but a collective action will create a bigger effect. It could urge corporations, governments, society, academia, media to undertake actions within a climate-conscious approach. As Wangari Maathai told us: "You cannot protect the environment unless you empower people, you inform them and you help them understand that these resources are their own, that they must protect them".



## Foreword 5: Fiona Alfionita Muda Karib



Currently, climate change has become an issue that has been discussed in every corner of the world. Climate change is one of today's most pressing issues in the world. As humans, we need to take part in dealing with this phenomenon. "Catalyst of Change: Navigating Climate Change Together" serves as a guide for us in dealing with this phenomenon. This book not only gives information but also covers the steps we need to take to safeguard our planet.

Climate change is a humanitarian problem, not just an environmental problem. This will have an impact on our daily lives. However, amidst the challenges we are facing, we can also see a glimmer of hope. "Catalyst of Change" demonstrates that if we all work together, we can bring about positive change that will have a significant influence on our planet.

In this book, we are urged to look at the issues that occur, as well as the remedies or activities that every individual must do, community and government because everyone serves as an agent of change in the fight against climate change. To accomplish better change, we must be able to join and combine our power and passion to effect big change, because every small or large effort we make will have meaning and an influence on climate change.

In this book chapter, we ask readers not only to gaze at the things that occur, but also to take action. We have the role to play in turning our earth into a better planet. With this book, we aspire to become change agents who will build a greener, fairer, and more sustainable world, ensuring a brighter future for future generations.

## Foreword 6: Muhammad Rizal



This book marks the inaugural endeavor by the Climate Catalysts team. Being the first of its kind, it has undoubtedly presented challenges at every stage, from conception and design to eventual publication. As a member of the Climate Catalysts team, I have actively contributed to this endeavor, particularly in assembling the core team responsible for bringing this book to fruition. Personally, I consider this book to be of paramount importance and relevance to readers, as it encapsulates insights from seasoned professionals and individuals deeply attuned to environmental and climate concerns. Furthermore, these insights extend beyond the boundaries of Indonesia, drawing from the perspectives of activists across various nations, thereby providing a comprehensive reference that broadens our understanding of environmental issues on a global scale.

I wholeheartedly recommend this book as an invaluable resource for enriching our understanding of climate and environmental matters. My aspiration for this book is that it receives widespread publication, allowing its message to reach a broader audience. I am confident in the exceptional quality of this book and believe that the knowledge it imparts will prove instrumental to readers, inspiring fresh perspectives and actionable ideas for fostering a healthier climate and environment. Clear conceptualization and effective implementation of these ideas hold the potential to translate them into tangible improvements, aligning with our noble objective of cultivating a society that thrives amidst favorable environmental conditions.



# Executive Summary

"Catalysts of Change: Navigating Climate Challenges Together" is a comprehensive initiative aimed at charting a course towards a diversified and sustainable future for our planet. Rooted in the imperative of preserving the health and resilience of our ecosystem, this program is designed to secure benefits not only for the present generation but also for those to come. By leveraging a wealth of best practices, innovative approaches, climate action projects, and scientific research, the initiative offers a multifaceted strategy to combat climate change from various perspectives. It brings together diverse stakeholders to collaborate on solutions that address the complex challenges posed by climate change.

One of the key features of "Catalysts of Change" is its emphasis on the involvement of young people from around the world. Recognizing the importance of fresh perspectives and innovative ideas, the initiative has actively engaged youth from countries such as Indonesia, Nigeria, Bangladesh, Iran, the United States, and the United Kingdom. These young voices bring a wealth of diversity and creativity to the table, enriching the discourse on climate resilience and sustainability. By empowering youth to take an active role in shaping the future, the initiative aligns with the objectives outlined in the Agenda 2030.

Through collaborative efforts and shared optimism, participants in "Catalysts of Change" are working hand in hand to realize a vision of climate resilience and a sustainable planet. By harnessing the collective power of diverse perspectives and innovative approaches, the initiative aims to drive meaningful progress towards a future where humanity and nature thrive in harmony. With an emphasis on concrete actions and outcomes,

"Catalysts of Change" is positioned to have a lasting impact on the global battle against climate change.

Overall, "Catalysts of Change" presents a unique opportunity to empower tomorrow's advocates and work towards achieving the UN Sustainable Development Goals at all levels, from grassroots to global. By fostering collaboration, innovation, and action, this initiative enables individuals and communities to play an active role in addressing climate change and promoting sustainable development. Whether it is through local grassroots initiatives or larger-scale global efforts, this book empowers people to make a positive impact and contribute to a more sustainable future for our planet. Together, we can work towards achieving the Sustainable Development Goals and creating a world where everyone can thrive within planetary boundaries.



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# CHAPTER

# 1

(**Authors :** Tanzeel Rashid, Hameedah Yahaya,  
Adamu Muhammad, Andi Farhan Fauzi,  
and Arya Fitrah.)



**Tanzeel Rashid (Bangladesh)**

Founder and President of Ogrograhi



## Unveiling Disparity in Disasters

In the ever-evolving landscape of climate risks, Bangladesh and Indonesia find themselves in the spotlight, grappling with the impacts highlighted by the 2021 Global Climate Risk Index (CRI). Ranked seventh and in the top-third, respectively, these countries not only share environmental vulnerabilities but also face a unique challenge – the disproportionate impact of climate disasters on women and girls.

In the face of disasters, women confront disproportionately higher risks of mortality and injury compared to men, largely due to entrenched gender inequalities. These disparities permeate aspects such as access to information, mobility, decision-making abilities, and resources. As a consequence, women and girls often find themselves struggling to access relief and assistance in the aftermath of disasters, hindering their recovery and perpetuating cycles of vulnerability to future calamities.

The harrowing experience of 25-year-old Shefali during the floods in Bangladesh vividly illustrates the challenges faced by women in such situations. "My sister gave birth in the midst of the water with no help. Pregnant mothers cannot swim this distance and often remain in unsafe houses."

Historical instances, such as the 1991 cyclone in Bangladesh, reveal stark statistics, with 90% of the fatalities being women and children. This tragic outcome was primarily attributed to women's inability to relocate to cyclone shelters while shouldering household responsibilities.

Similarly, the 2004 Tsunami brought to light a disproportionate impact on Acehnese women, who faced a death rate four times higher than men. The root cause was identified in the unpaid care work that kept them confined to their homes, while their husbands, many of them fishermen, were out at sea and unaffected by the deadly waves arriving onshore.

Recognizing these disparities, governments under the UN Framework Convention on Climate Change (UNFCCC) have adopted a Gender Action Plan. This plan champions women's active participation in climate processes and decision-making, emphasizing the need for a more inclusive and gender-sensitive approach to climate resilience.

But it's not just about policies; it's about empowering women. Beyond dismantling societal norms limiting women's mobility during disasters, it's about providing them with knowledge, skills, and resources to actively engage in climate solutions. This isn't just about equality; it's about crafting effective strategies for a resilient future.

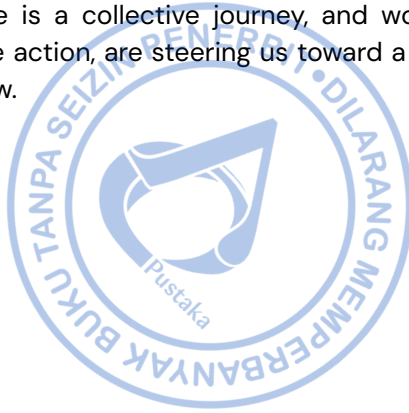
As climate challenges intensify, understanding and prioritizing the unique hurdles faced by women isn't just a call for equality – it's a vital step in creating comprehensive resilience. The



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path to a sustainable future is a collective journey, and women, leading the charge in climate action, are steering us toward a safer and more equitable tomorrow.





**Hameedah Yahaya (Nigeria)**

SUSI on Climate Change and the Environment Alumni 2023



## Nigeria's Environmental Promise

Nigeria, often hailed as the giant of Africa and the foremost oil producer on the continent, grapples with profound environmental challenges stemming from its extensive oil reserves. Despite its natural wealth, the nation faces a paradox: while endowed with abundant resources, it struggles to protect its environment and ensure sustainable development. This article delves into the intricate interplay between Nigeria's constitutional framework, environmental laws, and the practical realities on the ground.

At the heart of Nigeria's legal system lies the 1999 constitution, serving as the supreme law of the land. Within its provisions lies Chapter 2, which outlines fundamental objectives and principles, including the imperative to safeguard the environment.

Section 20 of Chapter 2 boldly states that. The state shall protect and improve the environment and safeguard the water,air, land, forests and wide life in Nigeria.



The implication of Section 20 of Chapter 2 is profound, as it underscores the constitutional recognition of environmental protection as paramount for Nigeria. This provision emphasizes the state's duty to safeguard water, air, land, forests, and wildlife, ensuring they remain healthy and conducive to human and non-human life.

Section 6 (6)(c) of the 1999 Constitution on the powers vested in the judiciary provides that these powers:

"shall not except as otherwise provided by this Constitution, extend to any issue or question as to whether any act of omission by any authority or person or as to whether any law or any judicial decision is in conformity with the Fundamental Objectives and Directive Principles of State Policy set out in Chapter II of this Constitution." The enforceability of this section is curtailed by Section 6(6)(c), which deems Chapter 2 non-justiciable.

This clause limits the judiciary's authority from adjudicating issues related to whether laws or actions align with the directive principles outlined in Chapter 2. Despite the preamble's assertion that the constitution is the product of the Nigerian people, the question arises as to why a segment beneficial to them would be made non-enforceable.

*"We the people of the federal republic of Nigeria....."* The preamble indeed highlights that the Nigerian people crafted the constitution. It raises a significant question regarding why the people would allow a portion intended for their benefit, serving as the backbone of the constitution, to be suspended. This contradiction prompts deeper reflection on the dynamics and complexities surrounding the enactment and implementation of constitutional provisions, especially when certain sections are rendered non-justiciable.

This constitutional conundrum also underscores fundamental issues of governance and accountability. Despite constitutional provisions ostensibly aimed at environmental protection, the practical implementation of these measures remains elusive. The disconnect between policy intent and execution raises concerns about the government's commitment to its citizens' environmental well-being.

In the realm of environmental awareness, Nigeria faces significant challenges. Many citizens lack basic knowledge about climate change, its causes, and its far-reaching implications. Bridging this knowledge gap is essential to fostering public engagement and galvanizing collective action to address environmental issues effectively. Moreover, raising awareness underscores the government's role as the custodian of public interests and environmental stewardship.

While environmental challenges loom large, there are opportunities for transformative change. Climate change, if approached strategically, can serve as a catalyst for innovation and sustainability. Initiatives aimed at repurposing waste into valuable resources demonstrate the potential for environmental resilience and economic growth. Imagine a future where everyday items like paper bags are used without detrimental effects on the environment, contributing to a more sustainable society.

In the realm of policy, Nigeria has made strides in enacting environmental legislation. For instance, the ban on styrofoam in Lagos and other states signals a commitment to curbing environmental degradation. However, the effectiveness of such measures hinges on robust enforcement mechanisms and governmental accountability. Despite legislative efforts, the persistence of environmental degradation in oil-producing communities like Ogoni underscores

the urgency of addressing systemic challenges and ensuring equitable environmental protection for all citizens.

In the case of *Wiwa vs Royal Dutch Shell*, Amnesty International described the executions as the culmination of a brutal campaign by Nigeria's military to suppress protests led by the Movement for the Survival of the Ogoni People (MOSOP), headed by Ken Saro-Wiwa. MOSOP asserted that the exploitation of oil resources in their land had enriched others while causing severe environmental degradation, turning their homeland into an ecological disaster due to pollution from spills and gas flaring.

During the early 1990s, Nigerian soldiers violently repressed the Ogoni people's protests against the operations of the oil company, employing deadly force and conducting massive raids. This repression peaked with the torture and execution of Ken Saro-Wiwa, a renowned writer and environmentalist, along with eight other leaders, collectively known as the "Ogoni Nine," on November 10, 1995. In seeking accountability for surviving victims and the families of those killed, legal action was taken against Shell in the United States.

In the realm of judicial activism, the landmark case of *Inakoju v. Adeleke* serves as a beacon of hope. The Supreme Court's intervention in challenging an ouster clause highlights the judiciary's pivotal role in upholding justice and accountability. The question arises: when will the Court similarly demonstrate activism in ensuring the justiciability of Chapter 2, thereby holding the government accountable for its environmental responsibilities?

Nigeria's environmental promise is contingent upon collective action, policy coherence, and judicial accountability. As the nation navigates the complexities of environmental governance, it must prioritize the alignment of constitutional ideals with practical realities on the ground. Only through concerted efforts

and sustained commitment can Nigeria fulfill its constitutional mandate to protect its environment and ensure a sustainable future for generations to come.





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## Catalysts of Change: Tackling the Climate Challenge Together

### Background

The unprecedented and alarming changes in Earth's climate system have generated serious concern among policymakers, researchers, and the general public over the devastating effects of climate change on people and the environment. Climate change has become one of the most pressing challenges bedeviling the world. Environmentalists believe that it is mainly caused by greenhouse emissions accumulated in the atmosphere by burning fuel, agriculture, and deforestation. According to the Intergovernmental Panel on Climate Change (2021), these among others lead to change in Earth's climate system, melting ice caps, rising temperatures, extreme weather conditions, biodiversity loss, sea level rise, and existential threat to human beings. To address this challenge, there is a need for collective efforts from all and sundry which requires

global cooperation, shifting toward sustainable practices, and adopting concrete strategies. Therefore, this chapter will delve into the best practices that will critically address the challenges of climate change and build a better future for all.

### **Best Practices for Addressing Climate Challenges**

To sustainably address the climate challenges, a multifaceted approach must be enacted which includes mitigation, adaptation, and building a resilient future. These best practices are discussed as follows:

#### **Decarbonization: A Cornerstone for Mitigation**

Decarbonization is primarily aimed at reducing greenhouse gasses through a transition from fossil fuels to renewable energy (REN21, 2023). Environmentalists such as Al Gore, Greta Thunberg, Bill McKibben, and Vandana Shiva, among others, argued that one of the best ways to mitigate the challenges of climate change and save the planet is to transition from fossil fuel to renewable energy sources like wind, solar energy, and geothermal power. As estimated by the International Renewable Energy Agency (2023), by 2050 renewables can contribute immensely towards addressing electricity demand by 86% which is one of the major sources of emissions. China, Denmark, and Costa Rica are some of the countries that demonstrated the feasibility of this transition, achieving significant penetration rates in this regard through Public Private Partnership and a strong policy framework in renewable energy (World Bank, 2022). Hence, by transitioning from fossil fuels to renewable energy we are contributing significantly toward addressing climate challenges and this can be done by everyone.

## Adaptation Strategies

Adaptation strategies simply mean adjusting and managing the inevitable effects of climate change. Although mitigation plays a vital role in addressing climate challenges, adaptation is also essential. According to the World Bank (2022), building flood defenses and seawalls, adopting early warning systems, and developing drought-resistant crops are essential practices for addressing climate challenges. In addition, adopting nature-based solutions like restoration of the ecosystem through reforestation initiatives plays a vital role in tackling climate challenges. It is widely believed that planting many trees helps significantly in reducing carbon footprint. According to research by the Swiss Federal Institute of Technology, planting a billion hectares of trees can store over two-thirds of the greenhouse emitted by humans since the industrial revolution. By embracing the planting of trees and refraining from deforestation, you are not only reducing your carbon footprint but also ensuring biodiversity conservation and the beauty of the planet.

## Building Resilient Future

Building a resilient future requires international cooperation and empowerment of individuals and local communities who are the major victims of climate change, making their empowerment and involvement very vital. The unprecedented existential threat of climate change against human beings and species necessitates international collective efforts, and solidarity among nations. The Paris Agreement of 2015 is one of the best agreements that provide a collective framework and seek global cooperation in tackling climate change across the globe. It set out a target for ambitious reduction of emissions and outlined financial and technological support needs for developing countries to ensure collective

efforts in tackling the climate challenges. Collective efforts and solidarity among nations will undoubtedly help in addressing climate challenges. To ensure this is archived, the empowerment of individuals and communities becomes very imperative. United Nations (2015) argued that supporting local communities with resources and building their capacity, engaging them in decision-making process, as well as supporting local adaptation initiatives are crucial for climate action. This will enable them to withstand future stresses and shocks. It is through empowering local communities to withstand climate challenges and contribute their quarters, ensuring strong social safety nets, and providing huge investment in upgrading our infrastructures (UNISDR, 2017). Therefore, to tackle climate challenges effectively, international cooperation and empowerment of individuals and communities are essential.

## **Conclusion**

Addressing the climate challenges demands a holistic approach that encompasses mitigation, adaptation, and building a resilient future. By adopting best practices in tackling the climate challenges such as building resilient infrastructures, embracing nature-based solutions, decarbonization, improving energy efficiency, fostering international cooperation, and empowering individuals and communities, we can successfully address the major challenges of climate change. To achieve this, it is necessary to ensure collective efforts where government, businesses, organizations, local communities, and individuals jointly become catalysts for change. As we embark on this transformative journey, let us remember what Mahatma Gandhi once said: “The future depends on what we do in the present”. Let us act urgently and collectively to save our planet. Remember that there is plan B but there is no planet B. Act now to save the planet for the present and build a climate-resilient future for generations to come.



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Universitas Hasanuddin



## **“Solution for the Environment, A Threat for Waste Pickers: Delving the Consequences of Implementing Deposit-Refund Scheme in Indonesia”**

The global climate crisis has become a major urgent event in the 21st century. The increase of greenhouse gasses, deforestation and exploitation of natural resources have contributed to accelerating climate change, which will eventually bring detrimental effects to the earth. This includes the increase of global temperature, the melting glaciers, the sea level rise, changing weather patterns, which all threaten the existence of humans and ecosystems. One of the important aspects of the climate crisis is the production and the disposal of waste, particularly the production of plastic which contributes to greenhouse gas emission, thus the waste requires hundreds to thousands of years to be decomposed which eventually



end up in the oceans, jeopardizing the marine ecosystem. According to the OECD (Ritchie, 2023), the emission from the plastic cycle including production and its disposal accounts to 1,8 billion tons. Given that the global emission counts to 54,6 billion ton CO<sub>2</sub>eq, the plastic is responsible for 3,3% global emission. To address this problem, various strategies have been proposed and implemented, including the deposit-refund scheme.

The deposit-refund scheme is a system designed to promote recycling by providing incentive to the consumer. By this scheme, the consumer paid a deposit included upon the purchase of bottled/canned beverages which will be returned if the packaging is stored for recycling. In November 2023, ten countries in Europe implemented the deposit-refund scheme including Croatia, Denmark, Estonia, Finland, Germany, Iceland, Lithuania, the Netherlands, Norway and Sweden. The advantage of the deposit-refund scheme is located upon the benefit to the producer and environment. The producers have no necessity to expedite huge costs for raw materials to produce packaging. In contrast, the producer recycles the packaging. On the environmental aspect, the deposit-refund scheme impacts the increase of recycling and decreasing demand of raw materials that simultaneously reduce carbon emission. The Zero Waste Scotland (2023) analyzes that the implementation of deposit-refund scheme may decrease the average emission of more that 160.000 ton CO<sub>2</sub>eq each year. This number is equal to the removal of 85.000 cars from roads, which demonstrates the significant impact of this scheme on carbon emission reduction.

Singapore is currently implementing the deposit-refund scheme to extend the age of TPS Semakau which became the only disposal place in the country. Even though Indonesia has implemented a deposit-refund scheme, one of the entities has

conducted a four-month study for implementation by providing incentive in the form of discount or cashback to consumers which returns the packaging (Vedithya, 2023). A survey by Amiruddin (2023) demonstrates that the interest of society to participate in the deposit refund scheme are majorly influenced by environmental awareness which leverages the collection of PET waste bottles in Indonesia. Although the reduction of plastic waste positively affects environmental quality and reduces carbon emission to address climate change, one of the issues is the fact that many Indonesian citizens tend to utilize waste collection as its primary source of income. The Waste Picker Society (IPI) (CNN, 2019) reports that 3,7 million of waste pickers in 25 provinces utilize plastic waste and other recyclables as income. This number may be increasing due to many provinces not included by the IPI.

The welfare of the waste-picker must become the primary consideration if the Government of Indonesia decides to implement the deposit-refund scheme. The increase of citizens returning bottles and cans to the designated site will simultaneously decrease the number of waste picked by the waste pickers. This situation is similar to the fisherfolks finding for other occupations if the number of catchments decrease. This issue may become serious if there are no available alternative solutions which potentially increase unemployment and criminal offenses. For example, Kompas TV (2024) reports that a waste picker in Semarang has been arrested due to stealing waterways metal cover. Therefore, a strategy which guarantees the welfare of waste pickers including empowerment and job field accommodation is highly needed to address the negative impact of the deposit-refund scheme.

This deposit-refund scheme offers the opportunity to increase recycling and reduce environmental pollution in Indonesia. However, this implementation scheme must be carefully performed

to ensure the economic and social impact, particularly to the waste picker must be accounted for. The waste picker plays a pivotal role in waste management in Indonesia and their welfare must become a priority, which is important for the government and business entities involved in deposit-return scheme to cooperate in initiating program and policies which support the continuity of waste picker including training and education, job creation and financial support. This deposit-return system is an important step towards a greener future. However, along with other policies, it is important to consider all effects and cooperate towards implementing a just and effective solution. With an effective approach, a cleaner and sustainable environment which also ascertains society's welfare is achievable.

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### **Background:**

The most dominant environmental issue in the last decade is global warming. In 2018, the United Nations (UN) issued a report in which scientists and government observers found the best way to avoid the worst climate impacts and maintain a livable climate for living beings by limiting the global temperature rise to 1.5°C. But with current trends in air pollution, global temperatures could rise as high as 4.4°C by the end of the century. The rise in the Earth's temperature can be caused by many things such as many human activities that are not environmentally friendly. However, did we know that based on the ISEAS–Yusof Ishak Institute report, Indonesia has the third highest position in Southeast Asia as a country that considers climate change issues serious and potentially threatening people's lives. We can be proud and appreciative for a moment, but the real actions of some Indonesians are still low in proving that they care about the climate. High awareness but not accompanied by real action, how can this happen?

## Discussion / Content:

Climate change still seems to be an invisible, slow-moving thing that doesn't really trigger people's awareness of the risks involved. That makes climate change a lower priority threat than more tangible threats like unemployment and crime. This is also in line with research from psychologist Robert Gifford on an attitude called "helping anxiety", where a person feels they don't need to do anything because someone else or some other group will do what is needed. In addition, many of us who are still concerned about the current condition of the earth give out appeals or warnings regarding the dangers and impacts of these abstract threats in the hope that people will take definite steps, but in reality it could be wrong because it might only make people afraid and not curious so that they will blame each other if a disaster occurs.

Setting a good example is the best way to inspire people to change. It would be better if we start taking action. For example, small things like limiting the use of air conditioners to reduce the release of CFCs into the air. Next, being conscious of the importance of using plastic products effectively, such as avoiding plastic bags when shopping and replacing them with reusable eco-friendly bags. In addition, recycling waste can help reduce the amount of waste sent to landfills, thereby reducing the pollution released when waste is incinerated. Reducing the use of private cars is also a very useful step as using public transportation or even walking can reduce the pollution that causes climate change.

The above steps may seem very simple and trivial, but if we continue to implement them consistently, it will have a huge positive impact on all of us in fighting the threat of climate change. As we continue to take simple actions that support the future of the earth and always invite the closest people, such as friends and family, hopefully they will participate. So big changes start from ourselves,

telling others and inspiring them to take real action together to protect the earth from the dangers of climate change.

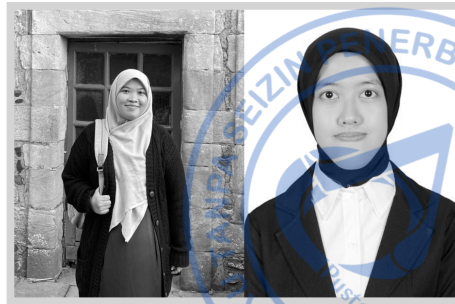
Climate change may seem to some people as a mere threat that may or may not happen, but what if it does happen and global temperatures rise to 4.4°C? Of course, we, the young generation of Indonesia, do not want our children and grandchildren to pay for this. It is better to pay the risk now than to pay dearly in the future. Therefore, by taking small actions, it is hoped that Indonesia will become one of the countries that can be free from climate threats earlier.



# CHAPTER 2



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Climate change creates a major effect towards health issues. Ironically, health service contributes significantly towards climate change. The primary cause of this climate change is the increase of greenhouse gas emission sourced from fossil fuel combustion including coal, oil and gas. According to the United Nations, fossil fuel is responsible for more than 75% global greenhouse gas emission and nearly 90% of total carbon dioxide emission.<sup>1</sup> Health Care Without Harm<sup>2</sup> in 2014 analyzed that the health sector contributes towards 4,4% of global emission or around two gigaton of carbon dioxide. This is equal to annual greenhouse gas emission resulting from 514 coal fired power plants.

The data from Carbon Brief in 2014<sup>3</sup> highlights that Indonesia was marked as the fourth biggest greenhouse gas emitter with 9,2% global contribution. Health sector in Indonesia contributes to 1,9% of national emission or equal to 13,6 million tons of carbon dioxide

and 71,6% emission proportion from domestic economic activity.<sup>4</sup> Although carbon emission from health sector tends to be lower than other sector, which is ironic, acknowledging that the health service commitment is the “first, do no harm” which translated to refraining from hazardous activity. Health service must become a recovery platform instead of disease burden.

Greenhouse gas emission in health sector originated mostly from fossil fuel energy consumption, particularly on electricity, gas and air conditioner combined with health sector operational.<sup>2</sup> Other source includes transportation, pharmacy and chemicals, waste management into agriculture (catering facilities, cotton plantation for medical attire, etc) which its production and distribution results to greenhouse gas emission.<sup>4</sup>

Similar to an infectious disease, environmentally unfriendly activities in the health sector will result in a serious and perpetrates prolonging health issue if remain untreated. The health sector is responsible to minimize its carbon footprint to net zero emission. This pathway includes various strategies and innovations which aims to increase energy efficiency, decreasing waste and promoting sustainability in facility and health operation.

In health operational sector, the health service in each level may apply the triple bottom line of quality improvement which highlights the achievement of individual and community health service on an economic, social and environmental sustainability framework.<sup>5</sup> Centre for Sustainable Healthcare (CSH) formulates the sustainable value, where a health service outcome must be achieved by minimizing impact or cost in economic and primarily on environmental aspect.<sup>5</sup> The seven elements of a climate conscious hospital according to the World Health Organization may become a solution to reduce carbon footprint in health services (Figure 1).

**Figure 1. Seven Elements of a Climate Friendly Hospital<sup>6</sup>**

1. Energy efficiency, which is to reduce the consumption and cost of energy use through efficiency and conservation efforts.
2. *Green building* design, which is building a hospital that is responsive to local climate conditions and optimizes energy and resource efficiency.
3. Use of alternative energy, i.e. using clean and renewable energy.
4. Transportation, i.e. using alternative fuels for healthcare vehicles, creating an atmosphere and facilities for staff, patients, and the community to walk, cycle, or use public transportation, and structuring healthcare areas that minimize patient transportation.
5. Food, i.e. providing food for patients and staff that is sourced from local, sustainable ingredients.
6. Waste, namely creating regulations and facilities to implement the 3Rs: *reduce, reuse, recycle*.
7. Water, focuses on conserving water use and minimizing bottled water consumption.

In addition, CSH also recommends the implementation of *Lean Service Delivery* principles (Figure 2), which is the implementation of health services that minimize waste in every procedure, process, and task through a continuous quality improvement system<sup>5,8</sup>. In applying these principles, everyone involved in health services makes efforts to reduce waste and eliminate everything that does not add value to patients.<sup>8</sup>

**Figure 2. Principles of *Lean Service Delivery*<sup>8</sup>**

1. Minimize patient waiting time.
2. Minimize inventory to reduce storage costs.
3. Eliminate defects to improve quality of care and increase *reimbursement*. Examples of defects are system failures, medical and diagnostic errors.
4. Reduce movement of patients, equipment, and materials to improve *patient flow*.
5. Prevent injuries and save time by reducing movement.
6. Maximizing resources and reducing overproduction of services, e.g. not using excessive tests/examinations, not prolonging hospital stays unnecessarily, etc.
7. Eliminate waste from redundant processing, e.g. by not filling in the same information on different sheets, inputting the same data on different systems, etc.
8. Understand that healthcare waste (processes and products that have no useful value) can hinder the development of the health sector's human resource potential, which is the culmination of 'waste' in healthcare.

Other solutions can be found at the community level. Innovations such as telemedicine or remote care models through communication technology can also help reduce the climate impact of healthcare<sup>3</sup>. By reducing the need for patients and health workers to travel to and from health facilities, telemedicine helps reduce carbon emissions generated by vehicles. In addition, telemedicine also supports the decentralization of health services especially in areas that have limited access to health services.<sup>3</sup>

Another aspect that is emphasized in creating *net zero* health services according to the *Sustainable Health Care Systems Model* is health promotion and disease prevention<sup>7</sup>. In this model, the focus of resource use in health care is no longer on curing diseases, but on how to foster healthy communities and communities that actively participate in building health. For example, the government and the community can work together to build green open spaces, increase pedestrian and cycling facilities, and build comfortable public transportation facilities. These efforts are made to motivate people to walk, cycle, and use public transportation for mobilization, which in turn can reduce individual and collective carbon footprints while increasing physical activity. Adequate physical activity can contribute to lower rates of non-communicable diseases such as heart and vascular diseases. Reduced use of private vehicles can also help reduce air pollution, which is an accelerator of the spread of airborne infectious diseases. If air pollution can be reduced, the spread of infectious diseases can also be reduced.

All efforts towards climate-friendly health services can only be realized if the health workers involved at every level have sufficient awareness and knowledge and skills to become agents of sustainability. The relationship between climate change and health, as well as the role of the health sector in creating economic, social and environmental sustainability need to be integrated in the education curriculum of prospective health workers and in the *continuing professional development* of all health workers.

Thus, sustainable healthcare can be achieved by jointly building *climate-smart* facilities, systems and human resources. By implementing climate-smart practices, the health sector can contribute to mitigating climate change and improving people's health and well-being. Given its mission to protect and improve health, the sector also has a special responsibility to implement the

Hippocratic Oath of "*first, do no harm*". However, it is not enough for the health sector, sustainable development efforts must also be integrated with other sectors. Without concerted action, the health sector will find itself at odds with increasing societal concerns about the social, political, economic and ecological dimensions of the climate crisis. Climate change, in all its dimensions, will be a top priority for decision-makers in every society.

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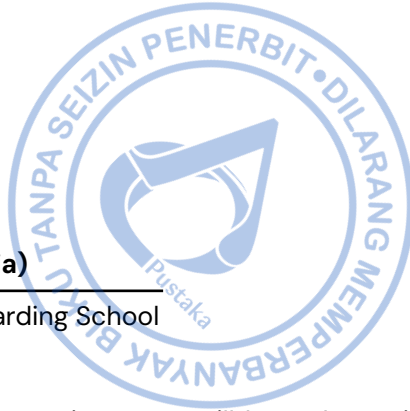


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Climate change in the coming years will be a determining factor for economic growth, as well as one of the main catalysts for policy-making and social change. Climate change refers to a change in climate status that is identified by changes in the mean and/or variability of climate-related factors and remains in effect for a broad or longer period (IPCC, 2007). According to The Economist Intelligence Unit (EIU) in 2018, Indonesia is one of the Southeast Asian countries most at risk of climate change. Indonesia as a country with diverse natural and food resources also determines the direction of policies that should be able to meet the needs and food security in a sovereign and independent manner.

According to Indonesian Law No. 18/2012 on Food, Article 1 paragraph 4 states that food security is a condition of food fulfillment for the state to individuals, which is reflected in the availability of sufficient food, both in quantity and quality, safe, diverse, nutritious, equitable, and affordable and does not conflict with religion, beliefs and culture of the community, to be able to live healthy, active and productive lives in a sustainable manner. However, climate change has an impact on food production, storage, availability, distribution and price stability (Nurhaliza, et al., 2023).

In addition, the impact on social change, such as inequality, welfare and prosperity of society. These conditions can be exacerbated by current lifestyles that tend to produce a lot of food waste (Food Wastage). According to Food Review Indonesia (2015), Food Wastage consists of Food Loss (food products discarded during post-harvest production, not reaching the consumer chain) and Food Waste (food products that are no longer wanted or discarded by consumers end up piling up in landfills).

Climate change triggered by human activities (anthropogenic) that produce "Food Wastage" as well as "Greenhouse Gas Emissions" increases global warming resulting in El Nino. El Nino is a phenomenon of warming sea surface temperatures above normal conditions which results in decreased rainfall. Food security relates to food crops grown under conditions of high atmospheric carbon dioxide leading to the loss of some essential nutrients. This is associated with malnutrition, which is an ongoing problem.

### **What is being done to face and tackle the problem of Food Security and Independence due to Climate Change?**

Climate change has many impacts on food security, such as decreased production due to failure due to long dry seasons or high rainfall, the spread of pests and plant diseases, government food reserve managers, supply and price instability. Therefore, it is necessary to optimize policies that are expected to realize food sovereignty through the availability and affordability of food consumption with nutrition and food safety both at the national and regional levels equally by utilizing local resources, institutions and culture.

Addressing malnutrition and hunger requires an integrated approach in accordance with the second point of the Sustainable Development Goals (SDGs), namely achieving food security, ending

hunger, improving nutrition through B2SA food, the Healthy Living Community Movement, and promoting sustainable agriculture.

Reducing crop loss or food loss through optimizing downstream reindustrialization and ensuring the reuse of food waste into more valuable products as well as preventive promotive actions of the Zero Food Wastage Movement.

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## Equity in Climate Crisis: Addressing the Disproportionate Impact on Vulnerable Nations

The disproportionate impact of climate change on countries that contribute the least to global warming represents a critical paradox in environmental policy. These nations, often characterized by lower industrial activity and smaller carbon footprints, find themselves on the frontline of climate change, including severe weather events, rising sea levels, and agricultural disruptions. This discussion examines the reasons behind this disparity and the role of international policy in addressing the issue by exploring environmental justice, global inequality, and the necessity for international cooperation.

The disparity arises from the accumulation of greenhouse gasses by industrialized nations, contrasted with the minimal emissions from many developing countries. These vulnerable nations,

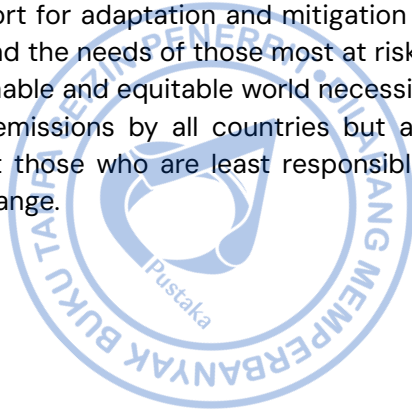
particularly in regions such as Sub-Saharan Africa, Southeast Asia, and the Pacific Islands are vulnerable due to their geographical positioning and economic dependencies on agriculture and natural resources, leading to a limited capacity for climate resilience and adaptation.

This vulnerability is not only an environmental concern but a problem that affects social, economic, and health sectors. For instance, the increased frequency of droughts and floods devastates local agriculture, threatening food security and livelihoods. Rising sea levels and extreme weather events displace communities, exacerbating poverty and social instability. The health impacts of climate change, including the spread of vector-borne diseases and heat-related illnesses, further strain already limited healthcare systems.

The principle of "common but differentiated responsibilities" supports the international policy framework on climate change, acknowledging the varying capabilities and contributions of countries to global emissions. However, the implementation of this principle has often fallen short of addressing the needs of the most vulnerable nations. Financial and technological support from developed countries is crucial for enabling adaptation and mitigation efforts in developing nations. Yet, international commitments, such as those outlined in the Paris Agreement, frequently encounter obstacles in mobilization, adequacy, and equitable distribution of resources.

The impact of climate change on countries with minimal contributions to global emissions highlights a significant injustice within the environmental crisis. This discussion has highlighted the critical need for adjusted international policy to effectively address the disparities in vulnerability and capacity. As we move forward, the global community must prioritize equity and justice in the climate

agenda, ensuring that support for adaptation and mitigation is on the scale of the challenge and the needs of those most at risk. The path towards a more sustainable and equitable world necessitates not only the reduction of emissions by all countries but also a collaborative effort to uplift those who are least responsible yet most affected by climate change.





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## **Relocation of the Capital City to IKN: Towards a Sustainable Future in Gender and Environmental Contexts**

Indonesia is currently entering a monumental era of transformation with plans to relocate the capital to the State Capital (IKN). This step is not just a geographical move but also a paradigm shift towards a more sustainable future. In this context, it is crucial to consider two critical aspects: gender and the environment. As a country committed to gender equality and environmental conservation, the relocation of the capital city must reflect these principles.

Firstly, let's examine the gender aspect in the capital relocation plan. Such changes often have different impacts on women and men. Therefore, it's important to ensure that every step considers the needs and contributions of both genders. For example, in planning new infrastructure, gender analysis is needed, considering women's



access to transportation, healthcare, and education. Economic opportunities resulting from new developments should also be equitable, considering women's economic interests as workers and entrepreneurs. By focusing on the gender perspective, the capital relocation can be an opportunity to enhance gender equality and strengthen women's roles in national development.

Secondly, the environmental aspect should also be a primary focus in the capital relocation plan. Large-scale infrastructure development often has negative impacts on the environment, such as deforestation, soil degradation, and air pollution. Therefore, a strong commitment is needed to ensure that the new capital's development adheres to environmental sustainability principles. Adopting environmentally friendly technologies in construction and transportation is one step, and preserving surrounding forests and wetlands is crucial for environmental conservation. Thus, the capital relocation becomes not only an opportunity to build modern infrastructure but also to reinforce Indonesia's commitment to environmental sustainability.

However, achieving a sustainable future vision requires active engagement from all parties, including the government, civil society, and the private sector. Active participation from various institutions and individuals will ensure that the capital relocation benefits not only economically but also considers social and environmental aspects.

In this context, an inclusive collaborative approach is key to success. The government should open up participatory spaces for the community, especially women and other vulnerable groups, to contribute to the planning and decision-making process. Additionally, the government should form partnerships with the private sector and international institutions to support project

implementation with the highest standards in gender equality and environmental preservation.

By considering gender and environmental aspects in the capital relocation plan, Indonesia can achieve an inclusive, sustainable, and highly competitive future vision. This step will not only physically transform the country but also reflect Indonesia's commitment to building a better future for all its people, leaving no one behind and respecting the natural heritage we possess.





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## Integrating Environmental Justice into “BreathWise” Education

### Overview:

Climate change has become one of the primary challenges faced by Indonesia in recent decades, referring to long-term shifts in temperature and weather patterns, influenced by natural factors such as solar cycles. This phenomenon is closely linked to the escalating air pollution in Indonesia. Global temperature rise can contribute to air pollution, particularly through increased energy usage for cooling and tropospheric ozone formation. Changes in weather patterns and extreme events like floods and forest fires can directly or indirectly exacerbate air pollution. Dust storms, vegetation changes, and biogenic emissions can also be influenced by climate change, affecting air quality. Global warming can further intensify transportation use, contributing to air pollutant emissions. The imbalances in ecosystems impacted by climate change can

affect the emission and absorption of air pollutants. In urban areas, rapid population growth and urbanization lead to increased motor vehicle and industrial emissions, contributing to air pollution. This air pollution poses an environmental problem with significant health impacts.

Various respiratory diseases are attributed to air pollution, with a high prevalence in Indonesia. Among the top 10 diseases per 100,000 population, four are respiratory diseases: Chronic Obstructive Pulmonary Disease (COPD) with 145 cases and 78.3 thousand deaths, lung cancer with 18 cases and 28.6 thousand deaths, pneumonia with 5,900 cases and 52.5 thousand deaths, and asthma with 504 cases and 27.6 thousand deaths. Thus, climate change and air pollution mutually reinforce each other, creating complex challenges that Indonesia must confront in preserving environmental quality and public health.

Climate change and air pollution reinforce each other, creating complex challenges for Indonesia in maintaining environmental quality and public health. BreatheWise emerges as a preventive solution to mitigate the risks of diseases caused by poor air quality. Dedicated as a contemporary web platform, BreatheWise monitors air quality in real-time through an IoT-based monitoring tool. The main dashboard presents clear and intuitive real-time air quality readings, empowering users to make informed decisions based on environmental information. This ensures users stay aware and proactive in safeguarding their health. The website also includes an informative section, educating users about the health impacts of various pollutants and offering practical tips to improve air quality.

### Goals:

- Provide accurate and real-time air quality information to users. This can help them understand the level of air pollution and its potential health impacts.
- Provide early warnings to users when the air quality around them decreases. The goal is for users to take precautionary measures as quickly as possible, so that BreatheWise can contribute to the improvement of users' quality of life and society as a whole.
- Provide users with historical air quality data so that they can track changes over time and understand air pollution trends in their region.
- Provide personalized health recommendations based on air pollution levels, helping users make better decisions to protect their health.
- To realize the highest degree of public health and reduce the incidence of diseases caused by air pollution.

### Potential for Ongoing Impact:

- *BreatheWise*, by building an active community of users, can increase community participation in environmental issues. This participation can create positive pressure for further action in environmental protection.
- The data collected by *BreatheWise* can provide valuable insights for the governments and environmental organizations to design better policies to address air pollution and other environmental issues.
- Improved public health resulting from the preventative measures informed by *BreatheWise* can have a positive impact on the economy, reducing the burden of healthcare costs and improving the economy, reducing the burden of health care costs and increasing labor productivity.







# CHAPTER

# 3

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**Refuji Zuma (Indonesia)**

Mentee of Climate Catalysts Fellowship Programme 2023



## Climate Change Adaptation and Mitigation with Biopore

### Overview:

Climate change is currently ongoing worldwide, particularly in Indonesia, and is attributed to greenhouse gases. These greenhouse gases include carbon dioxide (CO<sub>2</sub>), sulfur dioxide (SO<sub>2</sub>), nitrogen monoxide (NO), nitrogen dioxide (NO<sub>2</sub>), methane (CH<sub>4</sub>), and chlorofluorocarbons (CFCs). Among these gases, carbon dioxide (CO<sub>2</sub>) exhibits the most intensive movement into the atmosphere. Methane (CH<sub>4</sub>) is the second most crucial greenhouse gas, as it is stronger than carbon dioxide due to a larger radiative forcing per molecule. Additionally, methane absorbs radiation in an infrared window that is less saturated, allowing more molecules to fill that particular wavelength range.

Natural sources of methane include tropical and northern wetlands, methane-oxidizing bacteria that consume organic matter ingested by termites, volcanoes, seafloor seepage holes in regions rich in organic sediments, and methane hydrates trapped along the continental shelves in oceans and within the soil. Accumulation of organic waste that decomposes incompletely can result in methane gas production. Unmanaged organic waste generated by households has the potential to emit methane gas. The impacts of climate change on the environment and society can lead to floods, landslides, decreased soil fertility, reduced food production, and the emergence of diseases. Climate change is an inevitable occurrence in Indonesia and cannot be eliminated

Climate change is estimated to have occurred since the 1800s, with human activities emerging as a primary factor, especially since the industrial revolution. The use of fossil fuels such as coal, petroleum, and gas has significantly contributed to climate change. Despite technological advancements over time, many technologies developed have resulted in greenhouse gas emissions. To address climate change, adaptation and mitigation strategies are crucial. One solution is the use of biopores. Biopores are holes created in the soil with a depth of approximately 100–150 cm and a diameter of 50 cm, made from PVC pipes that have been perforated on each side. This innovative approach helps mitigate and adapt to climate change impacts.

### **Goals:**

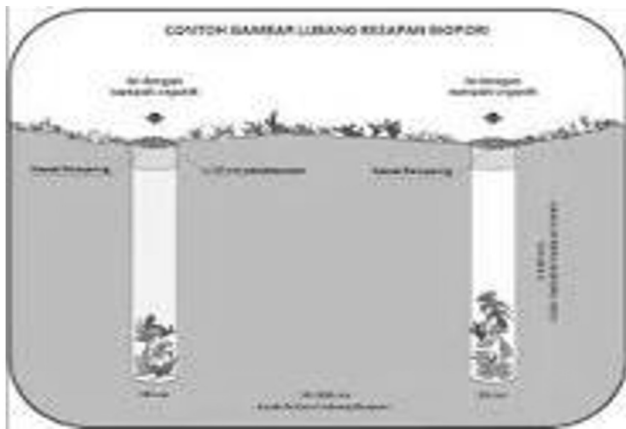
- To organize and manage organic waste on a household and large-scale basis in a systematic and flexible manner.
- To improve the quality of soil damaged by climate change by improving the physical properties of the soil so that the soil can maximize its function.



- To increase other water absorption spaces so that in the event of rain that changes in pattern and intensity can be overcome and also minimize the impact in the form of flooding

#### Potential for Ongoing Impact:

- The biopore holes that are designed for each residential area can be implemented by all people so that this is a real action for the social and natural environment.
- Densely populated housing is generally prone to flooding because the water catchment area is not conducive so that if la nina occurs it will be very detrimental to the community and here the function of biopori holes as a water catchment enhancer
- Creating healthy soil that is ready for climate change and creating small-scale agricultural spaces for communities so that if climate change impacts farmers who fail to harvest, the community can utilize their gardens.





**Wa Ode Sahara (Indonesia)**

Mentee of Climate Catalysts Fellowship Programme 2023



## **Sustainable Agriculture Development: Increase Small Farmer's Agriculture Productivity Use Farmer Friend (Farmend) App**

### **Overview:**

The impact of climate change such as high temperatures, and abnormal weather has resulted in many problems in the agricultural sector such as droughts, floods, pest and disease attacks, and others that can reduce farmer's income. But, agricultural activities especially in yield also produce emission gases. One of the solutions to solve the problem is education and technology. So, my climate action project aims to help small farmers in rural areas adapt and mitigate the impact of climate change in their agriculture area. Through Farmer Friend (Farmend) app provides a solution for small farmers to adapt and mitigate the impact of climate change



through collaboration with government and academics presented in a simple application that is easy to use by small farmers in rural areas. Agriculture students who are selected through selection based on several indicators such as CV and essay and passed will then collaborate through the Farmend application with the local agriculture institutions, Meteorological, Climatological and Geophysical Agency (BMKG), and farmers in rural areas to carry out online and offline services and services in the form of training related to land management to deal with the impact of climate change.

The features offered in the Farmend app are:

- **Course cultivate management:** The Farmend application presents information in the form of interactive training videos related to land management based on the implementation of Climate Smart Agriculture (CSA) to decrease emission gasses through enhancing carbon sequestration through innovative land-water management practice and implementation of Integrated Pest and Management (IPM) method which can be accessed online and offline.  
This course provides **environmental information:** related to the weather forecast and climate conditions to support agricultural activities obtained from the Meteorological, Climatological, and Geophysical Agency
- **Pest and disease detection:** this feature is based on Image Processing technology that can scan the problem. Through this feature, farmers can take photos of their plants, which will then display information related to the disease, type of pest attack, and control measures.
- **Course sustainable agriculture in climate change:** this course provides information about how to realize sustainable agriculture and the impact etc.

- **Horticulture plants:** farmers can also get information related to several horticulture commodities (vegetables and fruits) that are widely cultivated by farmers.
- **Free consultations:** farmers can get free online consultations from agricultural institutions and agricultural students, as well as online consultations that will be served by a team of agricultural students who will go directly to the field.
- **Community:** This feature can connect farmers between villages, allowing them to communicate about the condition of their farms, as well as setting planting schedules (crop rotation) as part of IPM implementation.

### Goals:

- To provide easily accessible and understandable courses about adapting and mitigating climate change for small farmers in rural areas
- To empower agricultural students through community service
- To develop sustainable agriculture besides Climate Smart Agriculture (CSA approach and Integrated Pest Management to decrease the impact of climate change.
- To optimize the relationship between farmers and the relevant government
- To maintain food security in extreme climate change
- To realize agri-food transformation systems towards green and climate-resilient practices

### Potential for Ongoing Impact:

- In Indonesian, many farmers come from rural areas, so the development of farmers in rural areas is important to develop sustainable agriculture from upstream to downstream



## CATALYST OF CHANGE Navigating Climate Change Together

- The Farmend app can increase public participation in climate change impact in the agricultural sector through students and local agricultural institutions' collaborations
- The data of pest and disease attacks on farmer land can be used as a reference for the agricultural local institutions in conducting counseling related to agricultural land management.
- The large number of agricultural student graduates in Indonesia can become a resource for developing the agricultural sector in Indonesia





**Ananda Ratu Azzarah (Indonesia)**

Mentee of Climate Catalysts Fellowship Programme 2023



## **AVENGERS : Training Greeners For Environmental Campaign Class | Environment**

### **Overview:**

It was recorded that 24.5% of 75% of the Gen Z population prefer Audio Visual Contents. However, currently the dominant campaign methods only use either audio or visual causing the message that would like to deliver through the content not provide cognitive effects, effective effects or even behavioral effects because the methods used are not appropriate with the current Gen Z. Furthermore, in considering numerous audiovisual contents on social media platforms but it does not captivate Gen Z's attention due to the delivery of the content editing lacks of interest. Therefore, I present the Climate Action Project which aims to enhance individual capabilities, particularly the videography skill of greeners in creating online campaigns regarding climate change



via social media platform, with an emphasis on video production. The activity agenda focuses on Coaching Class and Action Class which present experts in the field of videography and copywriting to provide capacity ranging from theory to implementation. This activity will continue regularly to enable greater involvement of Gen Z. Hence, it is hoped greeners Gen Z as mentee can act as Avengers to prevent Climate Change by spreading environmental information through social media platforms.

**Goals:**

- Enhancing the self-capacity (videography skill) of greeners to empower them in crafting online campaigns addressing climate change through social media with focusing on videography.
- Encouraging greeners to create online campaigns concerning climate change which are interesting and informative for Gen Z.
- Increasing public awareness about climate change through online campaigns created by greeners through video graphics.

**Potential for Ongoing Impact:**

1. The online campaign created by greeners has the potential to capture public interest and increase their awareness of climate change. It may be accomplished by creating engaging and informative video content, as well as using the appropriate social media platforms. By raising public awareness of climate change, society will get a greater understanding of its negative consequences and be more motivated to take action to mitigate it.
2. Effective online campaigns can encourage people to change their behavior to be more environmentally friendly. It can be achieved by providing clear and understandable information about the impact of climate change, as well as providing

alternative solutions that can be applied in everyday life. With a change in society's behavior, the greenhouse gas emissions that cause climate change can be reduced.

3. Creating agents of change Greeners who follow the videography mentoring activities will have the skills and knowledge needed to create an effective online campaign. Thus, they can become agents of change in an effort to prevent climate change. An agent of change from Gen Z, thus the campaign of climate change can be more effective and relevant to the target audience.





**Taskin Habiba (Bangladesh)**

SUSI on Climate Change and the Environment Alumni 2023



## Vertical Gardening: A Great Idea for Proper Urbanization

We have seen that Dhaka experienced the hottest day since 1965 where temperature soared to 42 degrees Celsius, making lives of over two-crore city dwellers miserable. And the mental and physical health with daily lifestyles have been impacted so much by this scorching heat.

And if we want to figure out the possible causes of this heatwave, the reduction of green spaces for decades is one of the significant reasons. Dhaka has lost around 56 percent of its green spaces over the last three decades. In 1989, this mega city had 12,745 hectares of vegetation, which came down to just 5,599 hectares in 2020, a new study has found. Population boom and unplanned urbanization are key reasons for the decline in green spaces. So, it is high time we took the initiative of green urban planning to save our lives from surface urban heat island intensity (SUHI). Again, we

have encountered the worst air pollution as well. So, these plants like English and devil's Ivy, chrysanthemums, lady palms – all work as natural air purifiers.

Thus, to combat climate change, urban forest is the cheapest and most sustainable solution to slow the magnitude of climate warming at regional to local scale. And vertical vegetation or gardening works as a blessing.

Vertical gardening or vegetation is an attempt to provide a greater crop yield per square meter of land. However, this term has been used to cover a broad range of approaches, from personal or community-scale vegetable and herb growing to vast skyscrapers for commercial production of a wide range of crops.

By 2050, the global population is predicted to reach 9.7 billion with 70% of people living in urban environments (UN,2015). For the expansion of infrastructure development, the concept of incorporating vertical gardening in the buildings is getting momentum for reducing heat islands. It is particularly suited to the cultivation of horticultural crops such as leafy vegetables. There are various ornamental plants which require less water than the traditional ones.

#### Low Water Outdoor Plants That Survive Summer Heat

1. Mexican feather grass
2. Blue oat grass
3. Purple fountain grass
4. Lavendula multifida
5. Cleveland sage
6. Bougainvillea
7. Morning glory
8. Money plant (devil's ivy, silver vine, hunter's robe)



9. Verbena
10. Echinacea

## Vertical Growth Surfaces

The facades of the buildings are proper green walls and it is found in a study that these facades are exposed to direct sunlight for a minimum of half the day, providing sufficient PAR for plants with high light requirements. The amount of available PAR (photosynthetically active radiation) increased with building height but was also influenced by façade orientation and configuration, with east-west oriented buildings considered to be better for continuous cultivation by avoiding the effects of north-south oscillation of the sun.

### Using Plastic Bottles For Gardening

Since plastic bottles are now a big issue to handle, we can use the plastic bottles instead of clay pots to plant trees. The plastic ones can be a durable solution to the traditional pots.

## Conclusion

The total urban population of Bangladesh is likely to reach 80–90 percent of the total population of the country in 2071. And to tackle this huge urbanization challenge, proper building design is a must now to tackle the climate change issues. The alarming part is we are already facing the severity of weather patterns.

In this circumstance, more and more greenery are recommended. So, the policy makers should put emphasis on green urban building planning. There should be a policy that every concrete building should have at-least a specific amount of greenery. And vertical gardening works as an immense advantage.



**Adi Indra Wahyudi Bahtiar (Indonesia)**

Universitas Hasanuddin



## Embracing Eco-Anxiety: Navigating Climate Change's Toll on Mental Health"

*"If you're often feel anxious and restless because of fears about natural disasters, environmental damage, or climate change doomsday, chill out! Your anxiety is valid, and you're not alone."*

The planet we inhabit is currently facing turbulent times. Natural disasters are rampant across various parts of the world. From deadly floods in Western Europe claiming over 200 lives to heatwaves in Turkey and Greece igniting forest fires in 2021. Indonesia also experiences various horrific natural disasters, data from the National Disaster Management Agency (BNPB) for the period 2019–2024 indicates that the most frequent natural disaster in Indonesia is floods, totaling 4,309 occurrences as happened prolonged floods in Sintang, West Kalimantan has submerged 12 sub-districts.



Despite the prevalence of floods, Indonesia also frequently experiences droughts due to unpredictable weather. It can be observed that drought disasters rank sixth, with a total of 220 cases. Although the number of cases is not as high as flood disasters, their impact is crucial for the community. This is because Indonesia is known as a country highly vulnerable to disasters. Indonesia's location amidst three tectonic plates has earned it the moniker "ring of fire," posing significant potential for natural disasters. These disasters are tangible examples of the impacts of climate change.

Climate change is believed to have adverse effects on our health due to increased air pollution, the spread of infectious diseases, and nutritional deficiencies resulting from food crises. Besides affecting physical health, climate change can also negatively impact our mental health. Concerns about the consequences of climate change can trigger excessive anxiety known as "eco-anxiety."

Eco-anxiety is not a medical term listed in the Diagnostic and Statistical Manual of Mental Disorders (DSM-5). However, in 2017, the American Psychiatric Association (APA) defined eco-anxiety as chronic fear of environmental doom. Eco-anxiety can be caused by various factors such as being at risk of natural disasters or environmental damage, negative media coverage, and feelings of guilt over the impact humans have on the Earth.

A study published two years ago in *The Lancet* looked at the issue of climate change and mental health based on a survey of 10,000 young people aged 16 to 25 living in 10 different countries. It found that almost 60 percent of young people felt "very" or "extremely" worried about climate change. More than 45 percent of respondents said their feelings about climate change affected their daily lives and the ability to sleep, eat, and study. Negative emotions expressed included feeling afraid, sad, anxious, angry, powerless,

helpless, and guilty.. This survey indicates that climate change can also impact our mental health conditions.

*So, what can we do to embrace eco-anxiety?*

1. Seeking Mental Health Services

Don't hesitate to reach out to professional mental health services such as psychologists. They can listen and assist us in coping with eco-anxiety or feelings of anxiety due to environmental damage.

2. Taking Action for the Planet

Engaging in conservation efforts can be done through our daily lives by adopting a more environmentally friendly lifestyle. Embracing an eco-conscious lifestyle aligned with our values and principles can help us live more mindfully. Some simple actions we can take include reducing plastic usage with reusable bags, bottles, and containers, sorting waste, opting for public transportation or walking instead of using private vehicles, and consuming more vegetables than meat to reduce carbon footprint.

3. Connecting with Nature

Spending time connecting with nature can provide us with several benefits such as improving attention, enhancing mood, reducing stress, and lowering the risk of mental disorders. Connecting with nature doesn't always have to involve visiting natural tourist destinations. We can visit city parks or bring greenery into our homes by gardening vegetables and planting ornamental plants in our yards.

4. Consuming Media Information Wisely

In the digital era, we are bombarded with a lot of information, including information about climate change.



However, consuming too much coverage of climate change-related natural disasters that make us worried is also not good for our mental health. Therefore, be wise in consuming media information and ensure that we always obtain information related to the environment or climate change from reliable sources.

On the other hand, the media also needs to consider showing optimism in environmental reporting instead of only reporting environmental damage phenomena in a fear-mongering way. This can be done by starting to cover various solutions and innovations available and feasible, thus fostering optimism in society. Educating the public about solutions and innovations to address climate change can also encourage people to take real actions for the environment.

#### 5. Advocating for Systemic Change

Although eco-anxiety is a personal matter, we need to realize that climate change is a structural and systemic issue. If we consider climate change to be a problem that can be solved solely by individual action, then we will tend to blame ourselves for the environmental damage around us. Therefore, we need to shift our focus of action for the planet towards structural and systemic changes. This can be done by building a collective movement to urge governments and private entities or producers to participate in and take responsibility for climate change issues.

Climate change is indeed an issue that needs our attention. The impact of climate change is predicted to be worse than the current pandemic. However, that does not mean there is nothing we can do to prevent and mitigate it. Currently, the Indonesian government has committed to reducing greenhouse gas emissions by 29% or 358 million tons of CO<sub>2</sub> through its own efforts and 41%

or 446 million tons of CO<sub>2</sub> with international support. As a society, we can also get involved in climate change prevention efforts both through individual efforts by adopting eco-friendly lifestyles and through collective efforts by joining environmental movements to advocate for more eco-friendly policies.

In conclusion, eco-anxiety, stemming from concerns about climate change, is a real issue affecting mental health globally. To address it effectively, we can seek professional mental health services, adopt eco-conscious lifestyles, spend time connecting with nature, consume media information wisely, and advocate for systemic changes. By taking these steps individually and collectively, we can work towards mitigating the impacts of climate change and fostering a more sustainable future for ourselves and future generations.







# CHAPTER 4



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**Aldiansyah (Indonesia)**

Universitas Hasanuddin



## Global Food Security: Sustainable Fisheries as a Solution in the Era of Climate Change

### Background

Food security issues have a significant global impact. According to a report by the Food and Agriculture Organization (FAO) of the United Nations, more than 820 million people in the world went hungry in 2019. Meanwhile, rapid climate change poses serious threats to food production and the sustainability of natural resources. In view of this, sustainable fisheries not only offer solutions to food shortages but can also maintain the balance of vulnerable marine ecosystems.

1. Sustainable fisheries, as an integral part of global food security, focuses on how to utilize marine resources without compromising the sustainability of marine ecosystems. With the increasing demand for fish as the main source of protein

for most of the world's population, responsible and sustainable fishing practices are becoming increasingly important. Involving practices such as maintenance of fish stocks, reduction of marine plastic waste, and preservation of marine habitats, sustainable fisheries face challenges and opportunities that require global attention.

2. Global food security and sustainable fisheries are not just issues of the agricultural and marine sectors, they also concern the well-being of humanity as a whole. In this context, addressing these issues requires collaboration between governments, international organizations, the private sector, and civil society. The relevance of sustainable fisheries lies in its contribution to meeting food needs, sustaining ecosystems, and addressing the impacts of climate change.

## **Discussion / Content**

Sustainable fisheries provide long-term solutions by ensuring fish resources are preserved. In the context of climate change, rising ocean temperatures and changes in marine ecosystems affect fish distribution. Therefore, fisheries sustainability helps manage fish stocks so that they remain sustainable. Data from the World Food and Agriculture Organization (FAO) shows that countries that implement sustainable fisheries practices tend to have larger and more stable fish stocks.<sup>3</sup> Sustainable fisheries support environmental sustainability. Environmentally friendly fishing practices help protect marine biodiversity and associated ecosystems. Analysis from the World Oceanic and Fisheries Research shows that the use of sustainable fishing methods, such as trolling and selective drift nets, can reduce negative impacts on the marine environment.<sup>4</sup> In analyzing the economic aspects of sustainable fisheries, it is important to consider that the fishing industry contributes significantly to the global economy.<sup>5</sup> However, ethical questions arise as the industry



grows. How far are we willing to sacrifice the environment and the sustainability of natural resources for economic gain?

Data from the UN presents a clearer picture of fisheries' contribution to global food. Around 3 billion people depend on fish as their main source of protein, while around 200 million workers are involved in the fisheries sector. Therefore, maintaining the sustainability of fisheries is essential to ensure global food security. Different views have emerged on addressing global food security through sustainable fisheries. Some argue that stricter regulations are needed to prevent the exploitation of marine resources, while others question their effectiveness. In encouraging critical discussion, it is important to recognize the complexity of this issue and seek balanced solutions.

## **Conclusion/Closing**

Fisheries sustainability provides assurance of preserved fish resources, in line with FAO findings showing that sustainable fisheries practices tend to result in larger and more stable fish stocks. The environmental aspect is also elevated, with environmentally friendly fishing practices helping to protect marine ecosystems and biodiversity. Around 3 billion people depend on fish as a primary source of protein, while 200 million workers are involved in the fisheries sector. Keeping fisheries sustainable is therefore essential to ensuring global food security. However, ethical questions arise regarding the growth of the fishing industry and the extent to which we are willing to sacrifice the environment for economic gain. This prompts critical thinking about the expansion of regulations and balanced solutions that can ensure the sustainability of natural resources without compromising economic growth. It is therefore necessary to take action in the face of recent climate change that could have serious impacts on national food security.

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**Nidia Mustika (Indonesia)**

Environmental Economics Enthusiast  
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## Green Economy: Why is it important, and How can we achieve this?

Indonesia possesses abundant natural resources. Utilising these potentials, Indonesia is capable of being a developed country whose citizens live a wealthy and prosperous life. Yet, it appears that this may not occur as expected.

Having studied economics, I became aware of the profound connection between the economy and the environment. Numerous economic activities harm the environment, and damaging the environment leads to substantial financial loss. I had witnessed firsthand some climate-related disasters in my hometown, such as floods, landslides, and a prolonged drought that resulted in hundreds of victims losing their earnings and lives. Following BNPB, at least 1,675 incidents from January 1 to May 31, 2023, were dominated

by hydrometeorological disasters at 99.1%. Without the proper measures, climate change-related disasters will likely become more intense and generate more massive harm. The Minister of Finance also remarked that climate change issues could adversely affect up to 3.45% of Indonesia's GDP by 2030.

The condition mentioned above is a critical challenge, considering that economic growth and environmental performance should go hand in hand. Any economic activities should be eco-friendly to ensure that the usage of existing natural resources can be enjoyed for future generations. Indonesia's determination to actively engage in the fight against environmental degradation is manifested in its participation in the Paris Agreement, which was developed in the spirit of climate change. Moreover, the government declares the green economy and low-carbon development the most suitable pathway to drive economic growth and improve social welfare while preserving environmental quality. According to the Ministry of Economic Coordination, the green economy has been embedded in the 2020–2024 RPJMN planning document, which is envisioned to be the means to pursue Indonesia's goal of reaching the world's top 5 economies by 2045 and of achieving a zero emissions goal by 2060.

Considering the state of affairs, it can be inferred that more environmental economists must be employed in a wide range of sectors and thus be more capable of supporting the realisation of a green economy in Indonesia. Through cost-benefit analysis, environmental valuation, carbon tax, and environmental subsidies, economics demonstrates a series of ways and means to resolve the ecological crisis.

One possible scenario to overcome environmental challenges is implementing a circular economy, for example, recycling unused products for longer use. This concept is currently

included in Agenda Prioritas Nasional. This model can promote the efficient use of natural resources and lessen the volume of waste disposed of. For example, the team of Sahabat Sampah community in Makassar (a community contributing to educating and promoting waste management) contributed to improving the environment by not only actively encouraging society to utilise all utensils wisely but also creating one innovation to distribute items that are no longer used to be donated to those in need and recycle them into other useful products. Not only does this minimise the quantity of domestic waste, but it also enhances the value of the goods.

Furthermore, following my discussion at EcoEdu.id, an agency that delivers workshops and talks related to the environment, I learned that cost-benefit analysis focusing on environmental indicators is urgently being accelerated in all areas. This situation is crucial given that a growing number of projects will be carried out in the coming years, and a proper assessment is required to weigh the risks of potential environmental damages and losses incurred. Another highlight is the importance of economic evaluation in formulating policies that reinforce sustainable natural resource management. Environmental valuation allows various parties to measure their resources' potential and the effects of economic activities, along with the financial implications of restoring environmental degradation.

In conclusion, a large number of experts from diverse fields are urgently needed with a common vision of dealing with the topic of climate change. Particularly, a growing number of environmental economists are required to address immediate gaps, needs, and policy priorities that can support the implementation of the green economy in Indonesia.



**Hadonia Lazarus Manurung (Indonesia)**

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## "Assessing Indonesia's Commitment to Coal Energy Use as a Follow-Up to the 2015 Paris Agreement"

### Background

The scientific study on climate change has been internationally recognized, coinciding with the formulation by the *International Panel on Climate Change* (IPCC), which asserts that human influence is the cause of a twofold increase in global surface temperature, rising by 0.74 degrees Celsius<sup>1</sup>. Specifically, climate change has had negative impacts on Indonesia. These impacts can be observed through the rising sea levels, which resulting in subsequent effects such as reduced land areas, damage to coastal ecosystem areas due to tidal waves, increased occurrences of extreme weather events such as El Niño heatwaves leading to



peatland fires, and La Niña wet waves contributing to periodic flooding<sup>2</sup>.

The condition mentioned above ignited a global discussion that has been compiled in the concept of *the United Nations Framework Convention on Climate Change* since 1994. This concept aims to achieve a stable state of greenhouse gas concentrations in the atmosphere at a level that is able to prevent anthropogenic interference which is dangerous for the sustainability of the global climate system<sup>1</sup>. The continuity of this concept is dedicated to protecting ecosystems, food production and sustainable economic development. Furthermore, this global document is elaborated in the *2015 Paris Agreement*.

*The Paris Agreement* reflects equality and the principle of shared responsibility, differentiated by the capabilities of participating countries. As stated in its clauses, this agreement is designed to restrain the increase in global average temperature from pre-industrial levels and to continue efforts to limit the temperature rise to 1.5 degrees Celsius above pre-industrial levels.

Assessing the consistency of participating countries in the Paris Agreement can be seen through the commitment of Indonesia. The Indonesian government, which has ratified the Climate Change Convention through Law Number 6 of 1994 about UNFCCC, is officially bound by obligations in the Paris Agreement. Indonesia's commitment to this global agenda can be seen from its target to reduce carbon emissions by 29% through national efforts and by 41% with international assistance by 2030. However, Indonesia's ideal steps are hindered by the harsh reality of its significant dependence on coal energy sources in maintaining the balance of national and international trade.

## Discussion:

Consistently, Indonesia still relies on coal to fulfill its electricity supply, with 56.1% of electricity generation coming from this industry. This situation is further exacerbated by the agenda of the Ministry of Energy and Mineral Resources, which plans to increase electricity production through coal by 15.7%, reaching 477.91 million tons compared to the previous target outlined in the National Medium-Term Development Plan (RPJMN) for the years of 2015 to 2019, which was 413 million tons. This is certainly not in line with Indonesia's commitment to the *Paris Agreement* through *Indonesia's Nationally Determined Contribution* (NDC).

Indonesia has developed a National Energy Policy aimed at achieving energy self-sufficiency and resilience. Therefore, in formulating the National General Energy Plan, the government not only manages national energy supply but also ensures that the allocated programs can have actual impacts on reducing greenhouse gas emissions. Referring to the current mitigation policies, it can be concluded that the policies implemented in the climate change mitigation agenda through the energy sector are outlined in the National Greenhouse Gas Action Plan (RAN-GRK). This includes increasing energy efficiency, using cleaner fuels, increasing the use of new and renewable energy sources, utilizing clean technologies for both power generation and transportation, and developing low-emission, sustainable, and environmentally friendly national mass transportation.

Based on the sources of emissions generated in the energy sector, especially electricity, the largest amount of consumption comes from the household sector which is condensed into two mitigation actions, namely core activities and supporting activities. Core activities may include increasing the energy efficiency in households and industries, developing low-emission power plants

using low-emission energy, and utilizing biogas for households. Meanwhile, supporting activities may include identifying geothermal potential data, developing new and renewable energy sources, and non-technical programs within the National Greenhouse Gas Action Plan (RAN-GRK) such as integration into the education sector, specifically the primary and secondary education curriculum.

The complexity of national electricity consumption requires the government to prioritize reducing the dependency on national coal production. This is based on the fact that the share of Renewable Energy Sources (RES) compared to the total accumulation of national energy use in the electricity sector is still relatively low, specifically at 10.5% of total production. Consequently, a significant portion of the energy used in power generation is still dominated by coal at 56.1%, followed by natural gas at 24.9%.

In a different context, a country's participation in international agendas brings forth an obligation to adhere to the consensus outcomes of those forums. This notion is further refined in the term "*pacta sunt servanda*," meaning agreements must be kept. As an international environmental regime, the *Paris Agreement* was crafted through various discussions, negotiations, and debates, resulting in the establishment of rules that countries worldwide must abide by. These preparations are projected to mitigate the impacts of climate change. The rules formed from these discussions then have implications for countries, encouraging them to join and implement them into national policies, this also applies to Indonesia. Indonesia, which has joined as a state party to the UNFCCC and has ratified the *Paris Agreement* through Law Number 16 of 2016, is obliged to fulfill the obligations contained in the approved agreement.

Further in this writing, the author aims to provide corrections regarding the implementation of the *2015 Paris Agreement* that

has been applied by Indonesia in securing the national coal supply. Indonesia's implementation focus in this sector is divided into three efforts, including policy coordination, policy coherence, and climate policy integration.

National coordination efforts are manifested in the form of the National Greenhouse Gas Action Plan (RAN-GRK) and the Regional Greenhouse Gas Action Plan (RAD-GRK), with the Ministry of Energy and Mineral Resources (ESDM) serving as the Coordinator. So far, mitigation efforts both at the national and regional levels are still being carried out through Presidential Regulation Number 61 of 2011, which was established in line with Indonesia's commitments before the Paris Agreement was ratified. This indicates that Indonesia's written commitments in its Nationally Determined Contribution (NDC) have not yet translated into mitigation efforts for climate change in Indonesia, particularly in the energy sector. If Indonesia continues to rely on existing scenarios for meeting its energy needs, it will be challenging for the country to fulfill its commitments to reduce greenhouse gas emissions by 29% through its own efforts and 41% with international support.

The second effort is policy coherence. Regarding this effort, the author did not encounter significant obstacles. This is because the sub-sector of energy provision with fossil fuels is excluded, except for natural gas. The final effort is climate policy integration. This effort is based on the characteristics and goals of climate policies, which aim to reduce emissions compared to business as usual scenarios. Therefore, transformation or changes are being carried out in this sector.

## Conclusion:

Based on the author's analysis, the impact of the Paris Agreement on Indonesia's policies, especially in the energy sector, has necessitated the integration of emission reduction goals outlined in Indonesia's Nationally Determined Contribution (NDC) with national energy policies. In the National General Energy Plan, the government has begun setting new final energy targets that increase the use of new and renewable energy sources, although in practice, the government has yet to clearly regulate policies regarding renewable energy. The author deeply regrets the lack of change or integration of Indonesia's commitments outlined in the NDC into efforts to mitigate climate change through the National and Regional Greenhouse Gas Action Plans (RAN/RAD GRK) in Indonesia.



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## Blue and Green: Women as Agents of Climate Change

In the current era, climate change determines our future. Climate change is a global challenge that requires collaboration from every element of society to achieve sustainable solutions in protecting the environmental ecosystem. Efforts to mitigate climate change can start with yourself. Especially us as women. The role of women appears as a crucial aspect that is often overlooked.

The colors blue and green, as the title suggests, create a visual image of the connection between women and the environment. Blue represents the sky and oceans which influence climate change. Good air quality affects health. On the other hand, the color green refers to forestry and agricultural ecosystems that

have great potential for environmental sustainability to maintain the environmental climate.

The trash that is usually found around us is of course plastic waste. Plastic is often found to be used to wrap food and drinks because it is considered more practical and the packaging price is cheaper. Apart from plastic, another problem that currently needs attention is the problem of waste of sanitary napkins and diapers or (diapers) used by women, babies, and parents. According to research by the Indonesian Consumers Foundation (YLKI), sanitary napkins containing chlorine can produce the chemical dioxin, which is a byproduct of the use of chlorine, generally used in paper products including sanitary napkins.

The Tampon Safety and Research Act of 1999 in the United States revealed that the effects of dioxin are cumulative and can remain in the body for up to 20 years after exposure. Although the law was not passed, the World Health Organization (WHO) categorized dioxin as one of the dirty dozen or dangerous chemicals which are persistent organic pollutants and carcinogens (cancer triggers). Then, another substance found in sanitary napkins and similar to dioxin is furan which is a paper bleach. Apart from health problems, disposable sanitary napkins and diapers also harm the environment.

The dioxin content comes from the Polychlorinated Dibenzo-P-Dioxin (PCDD) and furan comes from the Polychlorinated Dibenzofuran (PCDF) compound. The size of the compound is small and almost all compartments of the substance exist in the global ecosystem. The media for dioxin and furan distribution are very diverse, including air, water, and also through the food chain. In general, dioxins and furans can occur from low-temperature combustion. Burnt fuel can be solid and liquid waste, inorganic waste, which can be mixed with motor vehicle fumes, industrial fumes, forest fires, and

also cigarette smoke. Incomplete combustion of plastic waste will decompose as dioxin in the air.

Dioxins and furans are the result of burning domestic waste, medical waste, forest fires, smelters, recycling, power plants, factories, and others; even burning our daily rubbish. Incomplete combustion produces residual substances (residues) which should go through complete combustion at a temperature of 600 degrees Celsius. Diapers and sanitary napkins that are destroyed by burning will produce dioxin which is absorbed by plants and can then be consumed by animals and humans.

As previously explained, one of the consequences of damaged environmental conditions is the lack of creativity and care that befalls people when using these goods. Therefore, for waste management, contributions from the community are needed. Community concern is highly expected to become an agent of change in the form of participation and the need to revive environmental awareness. Community involvement in controlling waste on a household scale or domestic waste such as a sanitary napkin and diaper waste or (diapers) used by women to effectively, women's participation is required.

We can understand the role of women in protecting the environment as 'ecofeminism'. Ecofeminism is a movement that emerged among women in various parts of the world from various professions as a result of injustice towards women who are always associated with nature. Discussions about the environment are also related to ecofeminism as an implication of high feminist awareness among female scientists in universities in various parts of the world. Feminist women's awareness of the exploitation of nature makes them rise to play a role in saving the environment to create a better life eco-friendly and women-friendly. Why do women also need to contribute to saving the environment? Women are a pillar and role



model mainstay in a household. Women are considered to provide the first education to their children, with good understanding within women, they can provide education and understanding to all family members to care about the environment. A concrete example is when women who have become mothers and have children who are still babies will certainly need baby diapers for their children.

Baby diapers currently in circulation are still considered environmentally unfriendly because they contain dangerous ingredients that have an impact on health and the environment. In terms of safety, baby diapers in circulation sometimes cause rashes on babies' sensitive skin. Likewise, sanitary napkins that contain chlorine go through a process of bleaching or bleaching that is not good for the feminine area. Sanitary napkins containing dioxin can usually be recognized by the characteristics of burning and itching when used, causing vaginal discharge, an unpleasant odor, and a yellowish appearance. Currently, many sanitary napkins have been developed that are safe to use, such as those made from corn fiber, and bamboo fiber which better absorbs menstrual fluid, making organic sanitary napkins dry three times faster than non-organic sanitary napkins so they don't cause itching and are more hygienic.

Apart from that, reducing the waste of sanitary napkins and diapers can be developed into an innovation. For example making environmentally friendly sanitary napkins or diapers which was developed to make disposable sanitary napkins using more environmentally friendly materials, such as organic fiber, and reducing the use of plastic which is labeled "biodegradable" or "compostable", indicating that the waste can decompose naturally. This can also be done by recycling the sanitary napkins into planting media. Another innovation is modern cloth sanitary napkins which allow reuse with a design that can be washed and reused. This can also minimize the amount of waste produced by disposable sanitary napkins and diapers.

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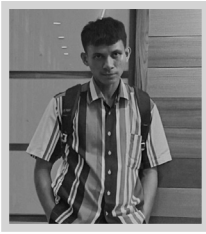




# CHAPTER

# 5

(**Authors** : Lion Marsyelyo Beay, Muhammad Chairul Sahar, Muhammad Maikudi Danburam, Nurul Rifqah Fahira, and Sartika.)



**Lion Marsyelyo Beay (Indonesia)**

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## Green Action & Experience with Alam Lestari Mimika

The year 2017 was the first time I became a student who dared to join a group of amazing friends. They were selected representatives from every Junior High School in Timika City, and yeah you're right... I was one of those representatives representing Yosua Timika Junior High School. Alam Lestari or abbreviated as "Allstars" is a program from the Environmental Department of PT Freeport Indonesia (PTFI) which in my opinion educates us with a basic introduction to the Environment and its problems, the 5Rs Rethink, Reduce, Reuse, Recycle, and Recovery were my favorite materials at that time.

The program lasted approximately six months, at which time we were the 11th batch and as I recall... we were fifty students representing each of our home schools. This was no ordinary program. Not only did we learn in every classroom in our home

school but we had the opportunity to socialize in the community. The program was quite complex, starting from receiving material, visiting workplaces that are part of the cooperation with PT Freeport Indonesia, and we even had time to carry out social-environmental services to one of the elementary schools in a remote area of Mimika Regency, namely Pomako One-Stop Elementary School.

We also showed our actions as students who care about the environment by participating in the routine cleaning of garbage on the roadsides in Timika city. On the other hand, we showcased clothes made from recycled waste (Recycle Fashion Show) which demonstrated the function of recycling waste with the simple goal of reducing the use of new raw materials.

The program also aims to find the Prince of Nature and Princess Lestari of Timika city from the quality and quantity of fifty students of this school. Prior to the selection there were several activities that we did, namely outbound in the MilePost 21 area, which is one of the tailings areas which is very beautiful scenery with chirping birds and lush Bintangur trees (*Calophyllum inophyllum*) that spoil my eyes and ears. At Mile 21 we really connected with the atmosphere of Nature and we even visited a building that displays a variety of butterflies, hence why I call it (Butterfly Garden).

Many experiences were very memorable and valuable when involved in the "Allstars" program but this is what I can describe in this short story. At that time I really felt that protecting and preserving nature is our responsibility as social beings. At the end of this program I was in the top 10 category and won the friendship category, this is what encouraged me to find friends who love discussing environmental issues and in 2023 "Climate Catalysts" is here to answer that urge.





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## "The Election of Denial: Celebrating over the Earth's cries."

The celebration of the five-yearly democratic feast has ended to choose a new leader who will continue the legacy of both executive and legislative power. The real count process to determine the winner is still ongoing even after the completion of all post-election stages. The euphoria among the lower layers of society has begun to decline as it becomes apparent who will occupy the seats of power for the next five years. However, along with the completion of this five-yearly democratic feast, there are still many things that leave a mark and must be reflected upon from the celebration that squanders hundreds of trillions of state funds. Many intrigues, gimmicks, phenomena, and so on, at least provide a glimpse of the 'backwardness' of society and the existing system. Examples include hate speech, hoaxes, money politics, and the chaos of billboards on the streets. In this article, let's take a brief look at the reflection of elections on their environmental impact.

## Campaign Props and Unethical Behavior

Throughout the year 2023 until February 2024, while traversing the streets in every corner of the city, street lights, and trees, there were rows of banners and billboards scattered everywhere, damaging the scenery. Although it has been regulated by the General Election Commission that Campaign Props (APK) should be installed considering ethics, aesthetics, cleanliness, and beauty, it seems that this is not adhered to by the election participants in an attempt to present themselves and be known among the community to pursue popularity and electability. It's not just a matter of visual comfort, but there are other issues haunting the earth as a result of this.

Advancements and developments in technology have not been properly seen as an opportunity by the candidates for the people's representatives, even though this year's elections are dominated by millennial and Gen Z voters, with each group comprising 66,822,389 (33.60%) and 46,800,161 (22.85%) voters, respectively, who are almost entirely connected to the digital world and social media. Campaigning in the digital world seems to be a new trend in this year's elections, but it seems to be merely a formality to appear to follow the progress of civilization. This has not been significant and massive by all election participants; they still rely on conventional banners and billboards. Sadly, such conventional methods have become a new environmental problem, as if we are celebrating democracy over the cries of the earth due to plastic carbon.

According to [timeforchange.org](https://timeforchange.org), the production of plastic banners requires 2 kg of crude oil and other materials for 1 kg of polyethylene plastic (PET/LDPE), and burning 1 kg of crude oil produces 3 kg of carbon dioxide (CO<sub>2</sub>). Therefore, for 1 kg of banner material, it contributes 6 kg of CO<sub>2</sub>, producing only 3 banners sized



1x1 meter (0.35 kg/banner). The contributed carbon may not be visible, but it is dangerous for the earth and life within it. Imagine the number of campaign props for candidates for the Indonesian House of Representatives (DPR) at 9,917 and the Regional Representatives Council (DPD) at 668, almost all of them creating campaign props, so how many million or even billion tons of CO<sub>2</sub> are contributed, not to mention the presidential candidates and candidates for provincial and district/city legislatures. As reported by Kompas.com, the CO<sub>2</sub> emissions from fossil fuel combustion in 2023 increased to 36.8 billion tons. This will have a significant impact on global warming and other environmental issues.

Aside from the environmentally unfriendly production of campaign props, their installation also sacrifices the trees on the streets. Many candidates for people's representatives install campaign props on trees using nails. According to walhisumbar.org, in December 2023, in the city of Padang alone, 2 kg of nails from campaign props were collected from trees. Nails driven by unscrupulous candidates into trees damage the tree bark and can obstruct water and nutrient transportation, thus affecting the growth and health of the tree and increasing the risk of disease infection in trees. This results in a decrease in the quality of trees or even the death of trees, leading to a reduction in the absorption of CO<sub>2</sub> and oxygen produced by the trees. This shows that one act of installing campaign props can lead to one of two serious environmental problems on earth.

This condition shows the environmental denial of the candidates for the people's representatives who claim to want to voice aspirations but instead damage the earth, which is the place of shelter for the people, for the sake of power desires. The election may have ended, but lessons must be learned for future democratic feasts so that we no longer celebrate over the cries of an

increasingly ailing earth. At least at the end of this year, we will once again anticipate the simultaneous regional elections throughout the provinces and districts/cities in Indonesia in 2024. The youth must voice out and be involved in this democratic euphoria, taking a centrist role in advocating for fundamental issues that are often overlooked by the general public.

Issues that are considered trivial but have a huge impact on all humanity, such as campaign props, will surely recur, so the youth must speak up and campaign for green elections with minimal plastic billboards and nails on trees. Becoming agents of change from conventional environmentally unfriendly campaign modes to modern campaigns through greener social media. Also, do not support political leaders who claim to fight for the people but actually damage the home (earth) of humanity. The youth are the largest segment of voters, so the struggles and voices of the youth will greatly influence Indonesia's greener future. Don't be apathetic because everyone will be affected by the environmental denial behavior of politicians. Preserve the earth as a blessing and trust from God to humanity. Sustainable greetings!

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# **New Technologies for Greener Shipping: The Mandate of Nigeria and International Maritime Organisation for Sustainable Shipping**

## **INTRODUCTION**

Can blue turn green? The right answer is a capital YES when it relates to the maritime industry. There is no questioning the fact that the oceans and other water bodies are crucial to life, perhaps the reason why God made them in abundance, occupying about 70% of the total earth's surface<sup>1</sup>. Not only do they pave way for economic breakthroughs by being the major route for international merchandise and the home for 80% of the planet's biodiversity, but also play a role in sustaining our ecosystem by absorbing about a quarter of the total global carbon output<sup>2</sup>.



Today, public discourse is centered around carbon emissions and climate change, the ultimate effect of which is global warming. Global warming is indeed a threat to the existence of life. The least it does is to cause the melting of glaciers in the sea that may result in flooding; the loss of the marine ecosystem; and the elevation of global temperature to a level incompatible with life<sup>3</sup>. This underscores the necessity for an urgent intervention for which the United Nations through its Sustainable Development Goals (SDGs) 9, 13, 14, 17, and ultimately the Paris Agreement (COP21) is committed to circumventing. The agreement was based on the goal of achieving zero emissions and stabilizing global warming at 1.5°C by 2050<sup>4</sup>.

Bound to the aforementioned industries, companies, and all individuals are mandated to come up with alternatives for a transition to cleaner, renewable, and affordable sources of energy in driving our industrial machinery. The shipping industry is one of those industries that can never be exempted from embracing this mandate as it accounts for more than 3% of the global CO<sub>2</sub> output which based on the current practices is predicted to increase to about 17% by mid of the 21st century<sup>5</sup>. Hence the obligation to innovate and deploy new technologies to fully harness the potential of the industry and at the same time ensure that the blue economy also turns green.

## **TECHNOLOGY: THE KEY DRIVER FOR SUSTAINABILITY IN THE MARITIME INDUSTRY**

"New Technologies for Greener Shipping" as succinctly captured in the theme of the International Maritime Organisation's 2022 World Maritime Day, depicts an advertent notion that underscores the fact that technology is pivotal to the shipping industry. Since the inception of the industry, the application of technology underpinned every facet of it, making shipping easier,

safer, and increasingly less costly. Indeed, the sustainability of the industry depends largely on advances in technology. This explains why many service companies and the International Maritime Organisation (IMO) commit huge resources annually to technological advancements in the shipping industry which makes it the backbone of the global economy accounting for about 90% of freight movement internationally<sup>6</sup>.

## **EARLY TECHNOLOGIES AND MODERNIZATION IN THE MARITIME INDUSTRY**

Early Technologies employed by the Egyptians, Austronesians, Chinese, and North America, among others, set the beginning of an industrial revolution in shipping, paving the way for innovations in the art of shipbuilding and navigation of the oceans. The primitive maritime technologies featured some small catamarans, outrigger boats, and lashed-lug boats that were used to traverse rivers by the ancient people, but as they begin to sail across the oceans the need for more power and speed necessitate the advent of new technologies<sup>7</sup>.

Ships were dually and multiply masted with huge sails to address the challenges of speed and force. Magnetic Compass was invented and used by the Chinese in orienteering, helping sailors to determine their direction of movement on water, and strengthening safety and security in the maritime industry<sup>8</sup>.

Fast forward to the beginning of the 19th century, ships were fully rigged and expanded, steam-powered engines were introduced, and wood was replaced by iron and steel<sup>9</sup>, enabling traders and explorers to carry as much tonnage of goods as possible and travel farther afield.

To this end, that shipping has been harmonious with the environment is indisputable, until the late 19th century when oil-



powered engines were introduced, they were more efficient and their usage increased dramatically<sup>10</sup>. Unfortunately for humanity, the duo became the major sources of pollution that drastically affected the environment which in turn threatened the existence of life. Thus for shipping companies to remain competitive, consolidating the existing technologies to innovate new eco-friendly ones becomes pertinent.

### **BEARING A TORCH TO LIGHT THE WAY**

The International Maritime Organisation (IMO) is providing leadership in technology advancement in the maritime industry, sparking interest and international consciousness by enacting regulations that drive innovations, promote collaboration and stimulate research and investment in critical areas such as shipbuilding, propulsion, pollution prevention, and energy-efficient technologies to cut greenhouse gases (GHG) emissions and foster greener shipping – notably through the establishment of MARPOL<sup>11</sup>.

### **NOTABLE ACHIEVEMENTS SO FAR**

Indeed, the torch is lit. With shipping companies shifting the paradigm to renewable sources of energy to power ships, deploying new technological solutions to upgrade performance, as well as showing commitments to sustainable partnerships; notable progress toward a greener future for shipping has been made; which include:

1. Electric powered and autonomous cargo ships<sup>12</sup>: e.g Yara Birkeland
2. Solar-powered fishing boats<sup>13</sup>: In its GHG reduction strategy, the Maritime Technology Cooperation Centre (MTCC) pioneered Solar-powered fishing boats for the fisherwomen of Nakalawaka village in Fiji.

3. Propeller Boss Cap Fins (PBCF)<sup>14</sup>: The MTCC is gearing the fabrication and installation of the PBCF on a Fijian Government ship to improve propulsion efficiency and reduce fuel consumption and associated GHG emissions.
4. AI for ship trim and optimization<sup>15</sup>: This is believed to significantly reduce fuel consumption and curb emissions as it saves up to 5% fuel during sailing.
5. Blockchain tech for data extraction and collection<sup>16</sup>: Software capable of collecting data on fuel and ship operations was used in Africa by MTCC and NIMASA to monitor GHG emissions from ships.
6. Biofouling prevention<sup>17</sup>: GloFouling project employs the use of robotics, antifouling components, and in-water cleaning systems to prevent Biofouling and protect maritime biodiversity.

These among other breakthroughs in the maritime industry sow the seed for cleaner and greener shipping, now and at any time in the future.

## **THE EFFORTS OF THE NIGERIAN MARITIME INDUSTRY**

Nigeria is a country squarely blessed both in human and natural resources. Nigeria's blue economy is vast and fertile, capable of generating over N7 trillion annually when effectively utilized. Despite challenges militating against the progress of the industry, NIMASA has never looked down on setting the ground for the deployment of technology and forging partnerships to protect the marine environment. The agency's commitments with MTCC, National Oil Spill Detection and Response Agency (NOSDRA), and the Maritime Environment Management (MEM)<sup>18</sup> are adequate determiners that NIMASA is applying multifarious approaches toward a greener future in shipping. It is therefore evident that we are also moving in the right direction.



turninggreen



## **OBSERVATIONS AND RECOMMENDATIONS**

Shipping is indeed a blessing to humanity and to sustain it we must be desirous to embrace changes and innovate new technologies that can prevent damage to our environment as well as the marine ecosystem. There is no gainsaying that the IMO and other local maritime bodies like NIMASA have been leading the way for a greener future in shipping in West Africa and the world at large, and to this end, so many grounds were shifted to the right. However, there is still the need for more especially in Africa where NIMASA is leading the change. Improvement of the maritime infrastructure, funding maritime technology initiatives, establishment of maritime centers of excellence in universities to align the academia with the industry needs, and adopting strong R&D frameworks will pave the way to speedily achieve our goals.

## **CONCLUSION**

That shipping has been around for centuries and has been making tangible impacts on the global economy is not under dispute. However, like many traditional industries, a willingness to embrace new technologies is crucial if shipping companies are to remain competitive. This therefore would necessitate that the public and the private sectors pool resources together to rejuvenate technology and innovation in the shipping industry.

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## Background

Food is a basic human necessity. Law Number 18 of 2012 concerning Food states that food must be available in sufficient quantities, safe, healthy, diverse, and affordable (Rosalia & Mulyaningsih, 2022). Food security is achieved when everyone, at all times, has physical and economic access to sufficient, safe, and nutritious food that meets their food needs and preferences for an active and healthy life (Rani & Reddy, 2023). Unfortunately, in Indonesia, as many as 74 districts/cities are recorded to have low food security due to constraints in availability, affordability, and utilization of food (BPN, 2023).

Amidst the trend of increasing food demand, worsening climate change also disrupts the food system. This is particularly evident in the context of global warming, which causes changes in rainfall and temperature. Changes in rainfall affect the volatility of dry and rainy seasons and alter planting seasons (Rosalia & Mulyaningsih, 2022). In tropical regions like Indonesia, rising temperatures increase the potential for pest attacks and the rate of water evaporation, thereby reducing production, especially of cereal crops (Firdaus et al., 2019; Rani & Reddy, 2023).

Until now, rice, corn, and wheat are still chosen as the cereals consumed by the majority of the Indonesian population. Unfortunately, these three commodities tend to be sensitive to extreme temperatures, so climate crises can reduce the quality and quantity of their production (Firdaus et al., 2019; Rani & Reddy, 2023). The community's dependence on the consumption of certain foodstuffs makes them more vulnerable to price fluctuations (Tranggono et al., 2023). Based on data from the Global Food and Security Index (2021), the extremely weak level of food diversity in Indonesia is one of the indicators causing food vulnerability issues in the country.

Food diversification can be an effort to transform the food system to achieve food security, facilitate access to healthy eating patterns, and improve nutrition (Soni et al., 2023). Indonesia has the potential to develop alternative carbohydrate sources with tropical cereals such as millet and barley (Nurmala, 2011). Efforts to increase the capacity of communities, both in terms of increasing knowledge and skills regarding alternative food commodities, are crucial to achieving community food security.

## **Discussion:**

Minor cereals are potential alternative food sources to address the climate crisis. Members of this cereal group are often overlooked, less popular, and have low consumption rates. Nevertheless, minor cereal grains are nutrient-dense foods with high bioavailability. Commodities such as barley, sorghum, and millet have great potential to be developed as food sources (Rodríguez et al., 2020).

### Jelai (*Coix lacryma-jobi*)

Jelai is a minor cereal plant with broad adaptability. This plant can be cultivated in both optimal and suboptimal land. Jelai's tolerance to water scarcity and resistance to pests and diseases is also better (Suarni et al., 2023).

As an alternative food source, jelai also offers diverse nutrients that align with modern dietary preferences. Jelai contains protein, fat, carbohydrates, calcium, phosphorus, iron, riboflavin, niacin, thiamine, fiber, and amylose. Jelai also has essential amino acids like methionine, which can act as antioxidants, boost immunity, and burn fat in the body (Suarni et al., 2023). Jelai's low glycemic index is suitable for consumption by diabetes patients, as it can reduce the rate of glucose absorption and suppress insulin secretion (Suloi, 2020). Various products such as rice, porridge, fermented cakes, and snacks like cookies have been produced from processed jelai (Juhaeti et al., 2021).

### Jewawut (*Setaria italica*)

Jewawut's potential as an alternative food in the midst of this climate crisis is demonstrated by its ability to adapt to marginal environments. Jewawut has a short life cycle and is self-pollinating. Compared to other cereal crops, jewawut has relatively high drought tolerance (Rodríguez et al., 2020).

Jewawut is a group of minor millets whose seeds are superior in protein content, fiber, phosphorus, vitamins B1, B3, and B5. Jewawut seeds also contain essential amino acids such as isoleucine, leucine, methionine, phenylalanine, threonine, tryptophan, and valine in higher amounts. Utilizing jewawut as a food ingredient is suitable for diabetes and cardiovascular patients due to its richness in beta-glucans and low glycemic index, which can improve sugar and cholesterol metabolism (Rodríguez et al.,



2020). Various minerals such as phosphorus, iron, and zinc can be found in jewawut. This millet also has higher levels of linoleic acid, tocopherol, phenols, and flavonoids, thus enhancing its therapeutic value (Pramitha et al., 2023). Several researchers have produced various innovations based on jewawut such as porridge, noodles, pasta, macaroni, nuggets, bread, cakes, dodol, chips, and traditional snacks (Ramlah et al., 2023; Susanto, 2022; Tzaniyah et al., 2023).

### **Milet Merah (*Eleusine coracana*)**

Red millet, often dubbed as "hunger food," becomes a candidate for emergency food due to its ability to grow in limited resources. This millet can withstand water scarcity in semi-arid climates due to its superior nitrogen use efficiency. Salinity stress can also be tolerated by this millet (Pramitha et al., 2023). The shelf life of red millet tends to be long, and it can withstand without significant damage from insect and pest attacks (Rodríguez et al., 2020).

The potential of red millet is supported by its nutrient-rich grains containing carbohydrates, proteins, minerals, and vitamins. This millet is rich in essential amino acids such as valine, phenylalanine, leucine, and histidine. Another quality of this millet is its gluten-free nature with a low glycemic index, suitable for celiac disease and diabetes patients (Rodríguez et al., 2020). The presence of threonine in this millet has the potential to inhibit fat formation in the liver and regulate cholesterol. The calcium, potassium, and iron content in this millet are beneficial for children during growth periods, breastfeeding women, and anemic patients (Pramitha et al., 2023).

Compared to other millets, the utilization of red millet as a food ingredient is still very limited in Indonesia. One innovation that utilizes this millet is as an instant porridge ingredient (Utami, 2022).

Instead of being processed as food, this millet is more commonly used as bird feed. Unlike in India, where this millet has been processed through milling, fermentation, and dehushing to produce products like noodles, soup, porridge, bread, and traditional cakes (Shobana et al., 2013).

### **Barnyard Millet (*Echinochloa* sp.)**

Barnyard millet has the potential as an alternative food due to its adaptability to biotic and abiotic environmental pressures. Its rapid growth is supported by its short life cycle (Renganathan et al., 2020). In addition to its agronomic advantages, this millet has a low glycemic index with high phosphorus, magnesium, dietary fiber, and niacin content. The high lysine and isoleucine content in this millet are beneficial for blood formation and metabolism. The high linoleic acid content is associated with antioxidant and plant immunological activities. Phytochemical compounds such as alkaloids, tannins, terpenoids, and flavonoids are also found in this millet (Pramitha et al., 2023).

Unfortunately, barnyard millet is still often overlooked despite its potential nutritional and agronomic value. In Indonesia, *Echinochloa* is still considered a weed (Tampubolon et al., 2019). The utilization of this millet is still limited as animal feed (Qomariah et al., 2014). However, India, one of the largest producers of this millet, has utilized it as a food ingredient processed into raw materials for bread, noodles, biscuits, biryani, laddu, and halwa (Renganathan et al., 2020; Verma et al., 2015).

### **White Millet (*Panicum miliaceum*)**

White millet is an ideal commodity for dry land with short growing seasons, shallow root patterns, and high water use efficiency. This millet is also tolerant of high temperatures, salinity,

and a wide pH range (Das et al., 2019). White millet has high protein, carbohydrate, and niacin content. White millet is also rich in essential amino acids such as lysine, leucine, isoleucine, and methionine. Its low glycemic index makes white millet a potential therapeutic agent for diabetes and cardiovascular diseases. This millet also plays a role as a source of minerals such as P, Ca, K, Na, Mg, Mn, Fe, and Zn (Das et al., 2019; Pramitha et al., 2023). The processing of this millet in Indonesia is still limited to snacks such as cookies and food bars (Anandito et al., 2016; Rianti & Retnaningsih, 2019).

## Conclusion:

The development of alternative food from commodities resistant to climate change is crucial for fostering food resilience in society. Jelai, jewawut, and other millets have great potential to be processed into nutrient-rich emergency food. These minor cereals need to be innovated to better align with societal preferences and introduced more widely to the public.

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## Opportunities and Challenges: International Climate Negotiations amidst Global Geopolitics

The issue of climate change is a central concern that attracts many countries to collaborate in solving the problems that trigger global climate change. Climate change can be an opportunity for countries to work together to mitigate global warming. The cooperation built through negotiations and diplomatic climate agreements is the result of a rationalization process involving all actors. In fostering cooperation, actors consider rationality, which involves weighing the benefits and costs for those involved in the collaboration. Against the backdrop of the current climate crisis, international climate negotiations are essential. The United Nations Framework Convention on Climate Change (UNFCCC) serves as a regime for the UN international organization. The Paris Agreement, one

of the UNFCCC agreements, is the result of diplomatic negotiations aimed at addressing climate change. The implementation of the Paris Agreement is hindered by intense geopolitical power politics related to historical contributions and dependence on fossil fuel expansion. One of the challenges in climate change negotiations is the difference in priorities.

Countries have different interests and priorities in facing climate change. Developed countries may be more focused on emission reductions, while developing countries pay more attention to economic development. Furthermore, challenges of equality and justice in international climate negotiations remain unresolved issues. Developed countries have made significant contributions to the accumulation of greenhouse gases in the atmosphere due to their industrial activities spanning centuries. Meanwhile, developing countries, despite increasing their emissions, still have much lower per capita emission rates.

Several meetings have not resulted in binding rules or rules dictating the achievement of desired goals, always depending on subsequent meetings to conclude agreements. In making an agreement, there are norms, rules, principles, and a decision-making process. Existing norms help limit the options of offers and demands from both developed and developing countries in negotiations and assist negotiators in formulating achievable and measurable demands. The interaction of principles and norms can be used to guide the demands of developing and developed countries in negotiating their share of climate change. This share can be divided into two schemes: developing countries are allowed to increase CO<sub>2</sub> concentrations to a certain level using a formula established in 2050, while developed countries are required to reduce emissions to a certain level with targets until 2050 influenced by five principles inherent in mitigation action plans, namely:

1. Equal burdens
2. Polluter pays
3. Modified polluter pays
4. Equal shares
5. Greenhouse development right

Every country in the world has its own policy to contribute to the issue of climate change. America and Europe themselves introduce subsidies, tax tariffs, and other policies aimed at accelerating the transition to environmentally friendly energy. This is due to political events indicating that the political momentum is shifting towards stricter national climate policies than before. Climate mitigation essentially has three components: funding for mitigation, adaptation, and compensation for losses and damages. Financing has been at the core of climate negotiations from the beginning. The Global Environment Facility (GEF) was launched as the first financing mechanism used to support the implementation of the UNFCCC, which includes other global environmental conventions such as biodiversity.

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# CHAPTER 6



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# Investigating the Role of Soil Microbial Communities in the Carbon Cycle and Climate Change

## Abstract

Anthropogenic development activities have resulted in rapid global climate change. The effects of increasing surface temperature are evident from unprecedented loss in plant and animal species. However, microbial life has not been well discussed in context of climate change. The present review has been done to understand the basic processes that soil microbial communities carry out in

context of climate change such as microbial mediated soil carbon sequestration and mitigation of greenhouse gases. Soil microbes play a crucial role in decomposition of organic matter and soil respiration processes that are fundamental in sequestering carbon in the soil and CO<sub>2</sub> emissions from the soil. Hence, an understanding of the role of soil microbes as both contributors to and reactive components of climate change is much needed to clarify their roles whether they can be used to mitigate the emissions of GHGs or accelerators of climate disaster at the macroscopic level and even at the global scale.

**We are the last generation that can do something about climate change!**

**Here, there is our mini review!**

Soil microbial communities play a crucial role in the carbon cycle, influencing carbon storage and cycling in terrestrial ecosystems (Lange et al., 2015). Understanding the role of soil microbial communities in the carbon cycle is essential for assessing terrestrial carbon cycle-climate feedbacks. Soil microbial ecology is complex and can be affected by climate change and other global changes, making it challenging to draw firm conclusions about the contributions of soil microbes to carbon exchange and global warming. It is crucial to consider both direct and indirect impacts of climate change on microorganisms and the complex interactions and feedbacks between microbes, plants, and their physical environment. Future research should focus on a multifactor experimental approach to better understand how soil microbes and their activities respond to climate change and their consequences for carbon cycle feedbacks (Bardgett et al., 2008).

Microbial diversity has been found to promote soil ecosystem functioning and carbon transformations (Maron et al., 2018). Manipulating the diversity of a soil microbial community can have significant effects on carbon cycling in soil (Maron et al., 2018). Microbial carbon use efficiency (CUE) is a central factor in regulating the flow of carbon through soil, but the interactions between biotic and abiotic factors that drive CUE are not well understood. Biotic factors, such as community composition and diversity, are strong predictors of CUE, while abiotic factors, such as temperature and moisture conditions, can modulate the relationship between diversity and CUE (Domeignoz-Horta et al., 2020). It is important to consider the multiple facets of global changes when studying changes in soil carbon cycling.

Soil microbial activity is a key driver of the carbon and nitrogen cycles and atmospheric trace gas turnover (Diamond et al., 2019). Understanding the metabolic capacities of soil microorganisms is crucial for understanding their role in carbon and nitrogen cycling. Genomic analysis of soil microbial communities can provide insights into their metabolic potential and the factors that influence their activity. Soil microbes are responsible for the processing of plant photosynthetic carbon inputs to soil, making them essential for the global carbon cycle. Changes in the soil carbon cycle can have significant impacts on atmospheric CO<sub>2</sub> concentrations (Gougoulas et al., 2014). However, current technologies lack the high-throughput ability to quantitatively apportion carbon use by phylogenetic groups and its use efficiency and destination within the microbial metabolome (Gougoulas et al., 2014).

More research is needed to better understand the microbial processing of plant-derived carbon and its implications for carbon sequestration and agricultural sustainability. Plant diversity and soil chemical properties are important factors that mediate the linkages

between soil microbial communities and ecosystem functioning (Beugnon et al., 2021). Plant diversity enhances the quality and quantity of resources for the soil microbial community, leading to increased soil carbon balance. Nitrogen fertilization has been found to weaken the linkage between soil carbon and microbial diversity, highlighting the importance of considering the effects of nutrient inputs on soil microbial communities (Yang et al., 2022).

In conclusion, soil microbial communities play a crucial role in the carbon cycle, influencing carbon storage and cycling in terrestrial ecosystems. Plant diversity has been found to increase rhizosphere carbon inputs into the microbial community, leading to increased microbial activity and carbon storage. Understanding the complex interactions between soil microbes, plants, and their environment is essential for assessing the contributions of soil microbial communities to carbon exchange and global warming. Future research should focus on a multifactor experimental approach to better understand how soil microbes and their activities respond to climate change and their consequences for carbon cycle feedback.





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## The Counterproductivity of Indonesia's Green Energy Initiative: The Case of Nickel Downstreaming

### Introduction

Energy has always been an interesting issue in economic development in various countries around the world. Growing populations and increasing energy needs so that countries continue to encourage the use of energy that is no longer controlled and then has an impact on the environment such as water pollution, air, and so on. Faced with the huge challenge of meeting a country's energy needs and still protecting the environment, in this context, green energy is a promising solution. Green energy, which generally refers to renewable energy resources such as solar, wind, water, and

biomass, offers great potential in various countries, especially for Indonesia.

Indonesia is one of the countries that has extraordinary natural resources including mineral resources which are very important in sustaining the Indonesian economy and has a lot of potential so that it is almost spread throughout the archipelago. Green energy as a window of opportunity for Indonesia to achieve progress as the government targets the renewable energy mix to reach 23% by 2025 and 44% by 2050 with various policies and regulations to support the presence of green energy in Indonesia (Katadata, 2019).

Looking further, Indonesia is rich in mineral resources that generate considerable income for the state through taxes and royalties every year, one of which is through nickel. According to the United States Geological Survey (USGS, ), Indonesia's nickel reserves are number one in the world, with total reserves of 21 million tons or 20% of the world's total reserves. Indonesia is also the world's largest nickel producer, with 1.6 million tons or nearly half of all world nickel production in 2022 (USGS, 2023).

Based on data from the Geological Agency mapping in July 2020, the Southeast Sulawesi, Central Sulawesi and North Maluku regions have the largest nickel ore resources in Indonesia (Ministry of Energy and Mineral Resources, 2020). The data shows that nickel natural resources in Indonesia have great potential to improve the country's economy through various ways, one of which is investment, this then becomes a method to develop wealth productively through the capital market.

The government itself has on various occasions been very open to the possibility of foreign investment in the nickel industry. However, some of the latest developments regarding the nickel industry, ranging from violations of SOP (Standard



Operating Procedure) in Morowali resulting in a smelter explosion to other impacts of facilitating nickel investment, tend to ignore other human aspects. This raises concerns that efforts towards a green energy transition in Indonesia could be counterproductive in its implementation. The author in this paper tries to provide an explanation of how efforts to bring green energy in Indonesia in the form of nickel downstreaming tend to be counterproductive.

## **Discussion**

### **Elites and Dependency in the Politics of Nickel Investment in Indonesia**

Nickel is an abundant resource in Indonesia. Seeing this, the government sees an opportunity in its utilization. Processed nickel products can be useful in many ways, especially as a raw material for batteries for electric vehicles which have recently experienced a very strong trend. To manage these nickel resources, the government needs human resources in the form of capital and experts, which Indonesia currently does not have in large numbers. Therefore, the government has recently been very aggressive in promoting nickel downstreaming (Islamiati, 2023).

For this reason, the government has taken several policy steps that make it easier for businesses to enter Indonesia. Several government policies such as the establishment of the Job Creation Law, the Revision of the Minerba Law, and the National Strategic Project (PSN) were intensified. Unfortunately, these policies are loaded with an exploitative authoritarianism agenda (Tasmalinda, 2024). The policies are launched unilaterally without any community involvement in both policy formulation and implementation.

The government has turned a blind eye to all the land grabs, health problems and environmental damage that have occurred in

the name of investment. As for the investment concessions given by Indonesia, many Chinese companies came to invest in Indonesia. Unfortunately, Chinese investment in Indonesia has created a very strong investment dependency. With the intensification of the downstream program, investment from China (including Hong Kong) has increased sharply in the last 10 years from only 5.1% of total foreign investment in 2014 to almost 30% of all foreign investment in Indonesia (Singgih, 2024). This economic dependence is dangerous, given that if there is an economic shock from China, it will disrupt the Indonesian economy. Especially when several economic projections from international institutions such as the World Bank mention the decline in China's economic growth as well as the property crisis in the bamboo curtain country there (IDX Channel, 2024) which began with the bankruptcy of property giant Evergrande (Arbar, 2023).

### **The Impact of Nickel Downstreaming in Indonesia**

Downstreaming is the process of processing raw materials into semi-finished or finished products that have a higher selling value to provide added value to a product by carrying out a series of long-term oriented advanced processing processes. Downstreaming carried out in Indonesia is downstreaming that transfers technology, utilizes new and renewable energy sources, and minimizes environmental impacts. One of them is nickel downstreaming which is part of the industrialization process that encourages the transformation process from a country with a high level of dependence on its natural resources to an economically independent country for the prosperity of the people. The success of domestic industrialization cannot occur only with the support of the downstream mining sector, but must also be supported by industrial development that is more down-to-earth touching community participation.

One of the largest downstream investments in Indonesia is concentrated in Morowali Regency with the establishment of the Indonesia Morowali Industrial Park (IMIP). When the latest case is the smelter explosion, it raises concerns about other impacts of nickel investment facilitation that tend to ignore other human aspects (Affan, 2024). This raises concerns that efforts towards a green energy transition in Indonesia could be counterproductive in its implementation.

The mission of supporting the conversion of emission-intensive energy to green energy at the expense of environmental conditions is paid for by high deforestation and the destruction of marine ecosystems is not worth it. The real impact of waste disposal is the sedimentation of the Morowali coast. Morowali's mangrove forests and coastal seas, which are home to fisheries resources, are damaged and have resulted in a decrease in income for fishermen. Fishermen now have to go further out to sea to catch fish. Nickel mining activities have had an impact on Morowali's maritime environment (Syarifuddin, 2022). Every mining plan should consider waste management and the environmental impacts it may cause so that balance is maintained.

## **Conclusion**

This analysis highlights the contradiction between Indonesia's efforts to introduce green energy through nickel downstreaming and the resulting negative impacts. Although green energy is considered a promising solution in addressing environmental challenges and meeting energy needs, the implementation of nickel downstreaming tends to be counterproductive. This is due to the imbalance between the interests of political and economic elites and the aspirations and needs of the people. While the government is pursuing policies to attract foreign investment in the nickel

industry through downstreaming, the impact on local communities and the environment is often ignored. Violations of operational standards and serious environmental impacts, such as smelter explosions and deforestation, show that the policy favors investor interests over environmental sustainability and community welfare. There is therefore a need for a change in policy approach, ensuring public participation, transparency, and an emphasis on sustainable and equitable development for all parties involved. Only with a comprehensive approach that is based on the needs of the people and environmental protection can Indonesia achieve a transition to green energy without compromising welfare and the environment.

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The effects of climate change on children. Climate change affects everyone. Babies and children are the most affected. Heat can cause kidney disease and heat stress for children and endanger their mental health. Climate changes such as floods and storms are childhood injuries. Climate changes affect children in poor countries much more.

Health problems, infant death, and asthma are among the effects of climate change on children. Placing children and experiencing climate shocks, droughts, floods, affects children's growth, and their progress and learning will have a bad effect. These negative consequences affecting about one billion children in the whole world is threatening.

Two-thirds of the world's countries do not prioritize children in their climate plans. Climate change is usually caused by the burning of fossil fuels, which causes air pollution and the emission of greenhouse gases. It has a very destructive effect on children and causes breathing problems in them. Use more clean energy sources. Global warming has many effects on children's lifespan and quality of life. A child who is born today will be in a world that is four degrees hotter when he is 71 years old, and his health is severely threatened.